

# Decrypting Blockchain

THE ART OF THE STEAL



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- 1 PSEUDONYMITY
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- 3 SAVE THE KIDS



# Blockchain Fundamentals

# Hash Functions

1.1



Plain Text



Hash Function



```
#bc1!dn  
1 s & " ( #  
g d # ! @  
1 % $ G h  
9 - 0 # $
```

Hash Value

**4A7EF1BC**

arbitrary-length,  
long input



**Hash  
Function**



**47AE**

fixed-length,  
short output

4A7EF1BC



**Hash  
Function**



47AE

DC97840F

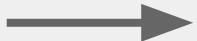


**Hash  
Function**

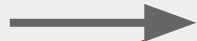


47AE

4A7EF1BC

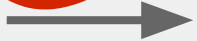


Hash  
Function

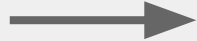


47AE

4A7EF1BC



Hash  
Function



47AE

**THIS IS HARD.**

**Double  
SHA-256**



**Keccak-  
256 Hash**



# Elliptic Curve Cryptography

1.2

ECC Encryption

Size: 256 bit



Same Security

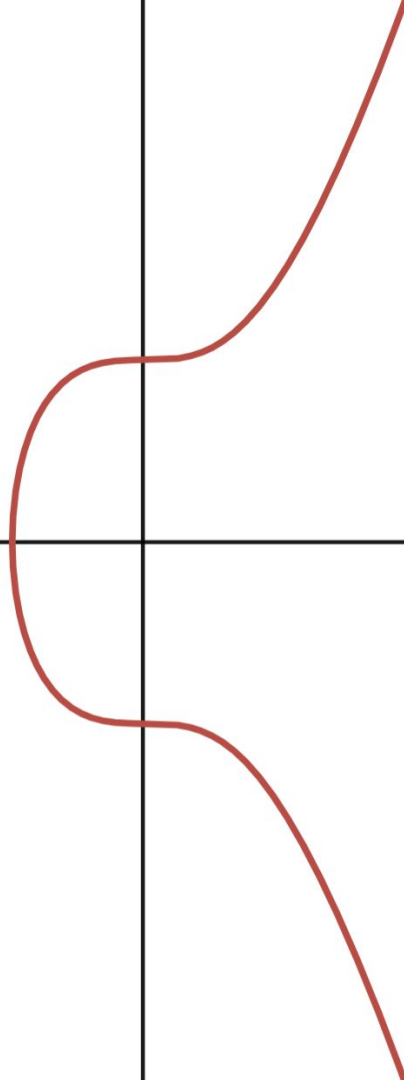


RSA Encryption

Size: 3072 bit

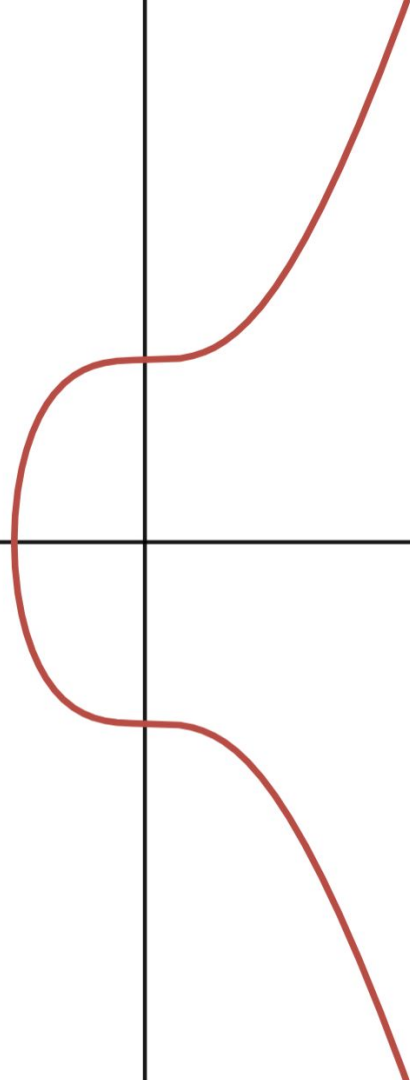
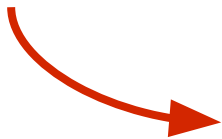


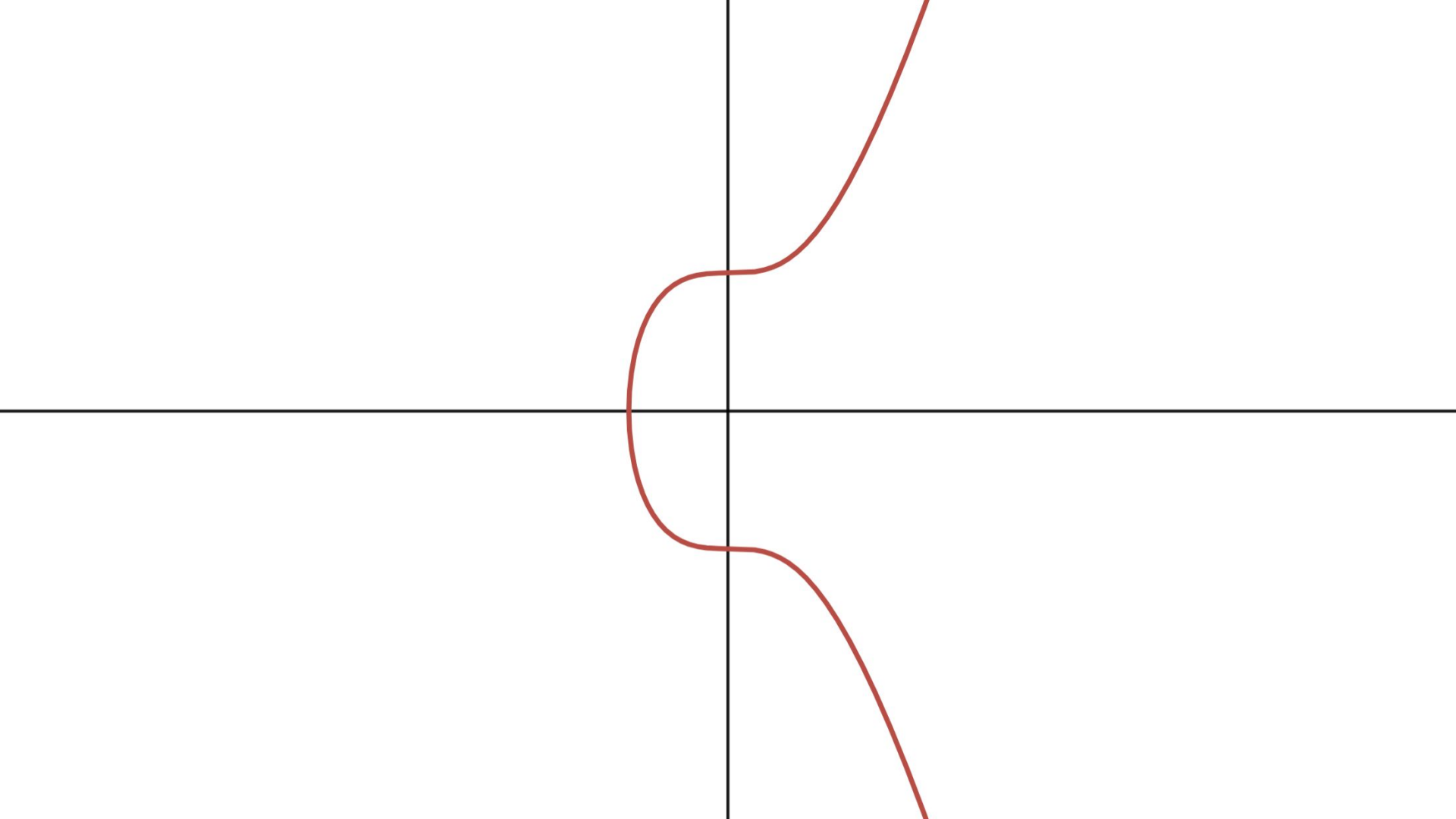
$$y^2 = x^3 + ax + b$$

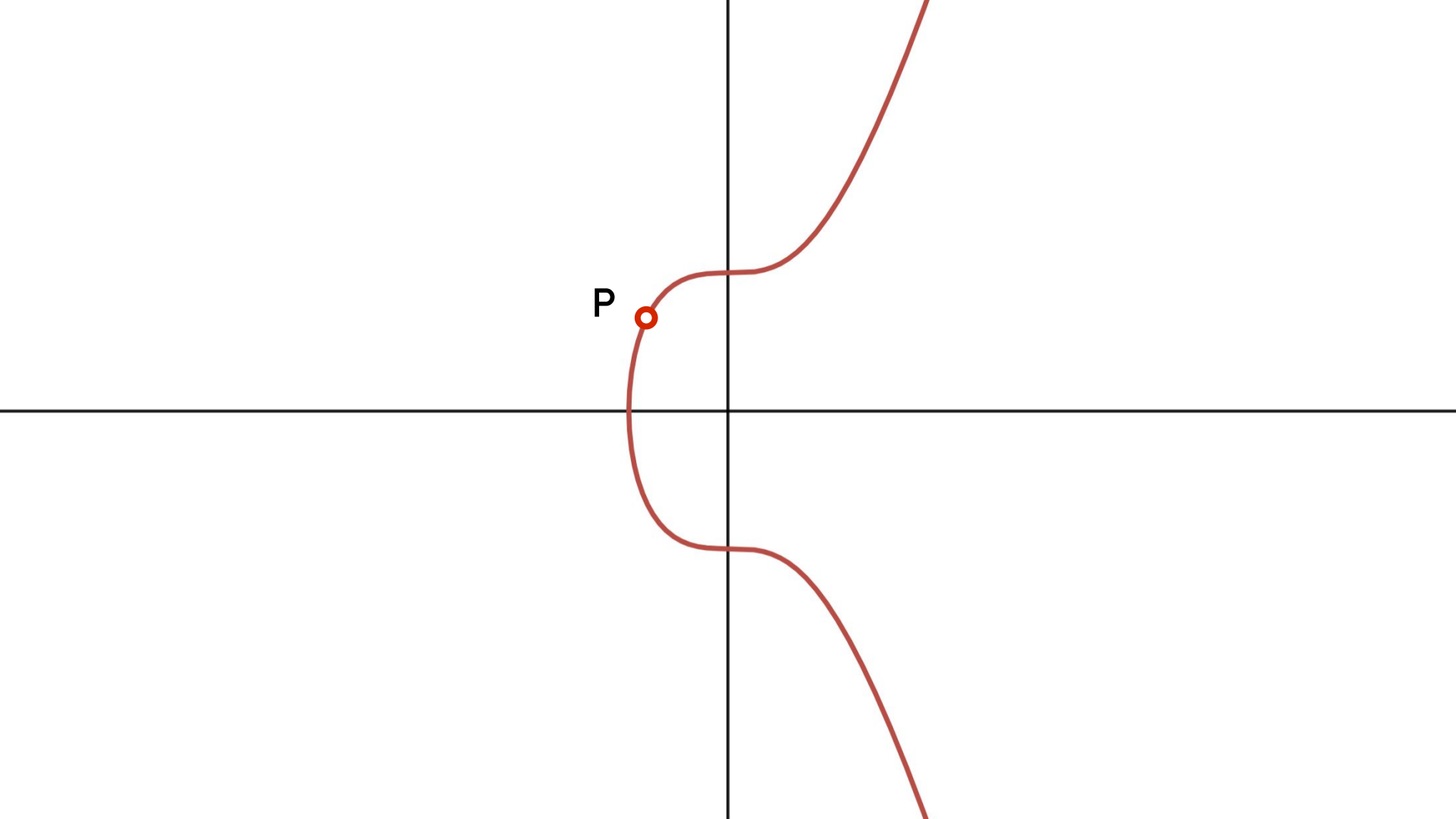


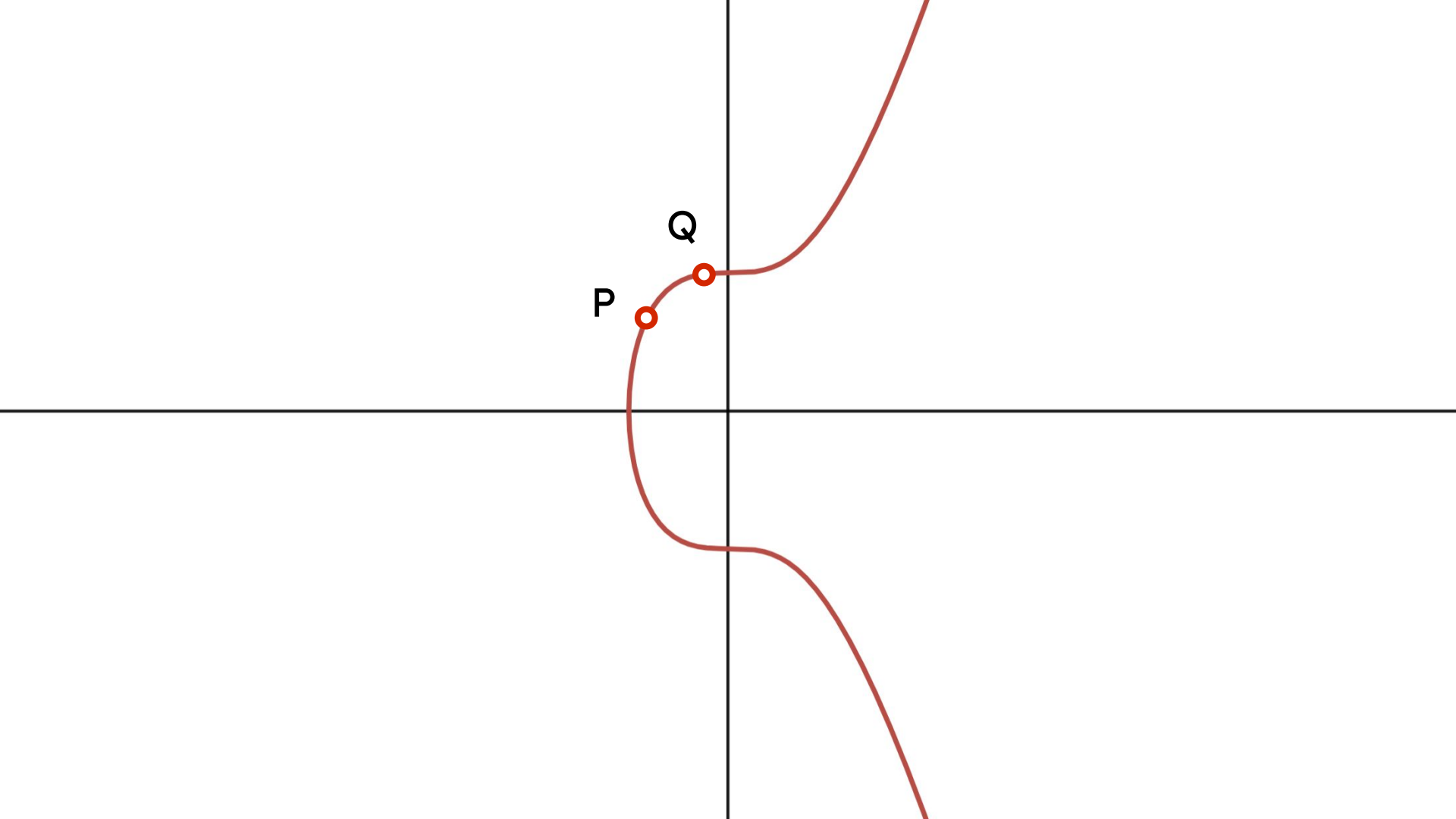


$$y^2 = x^3 + 7$$





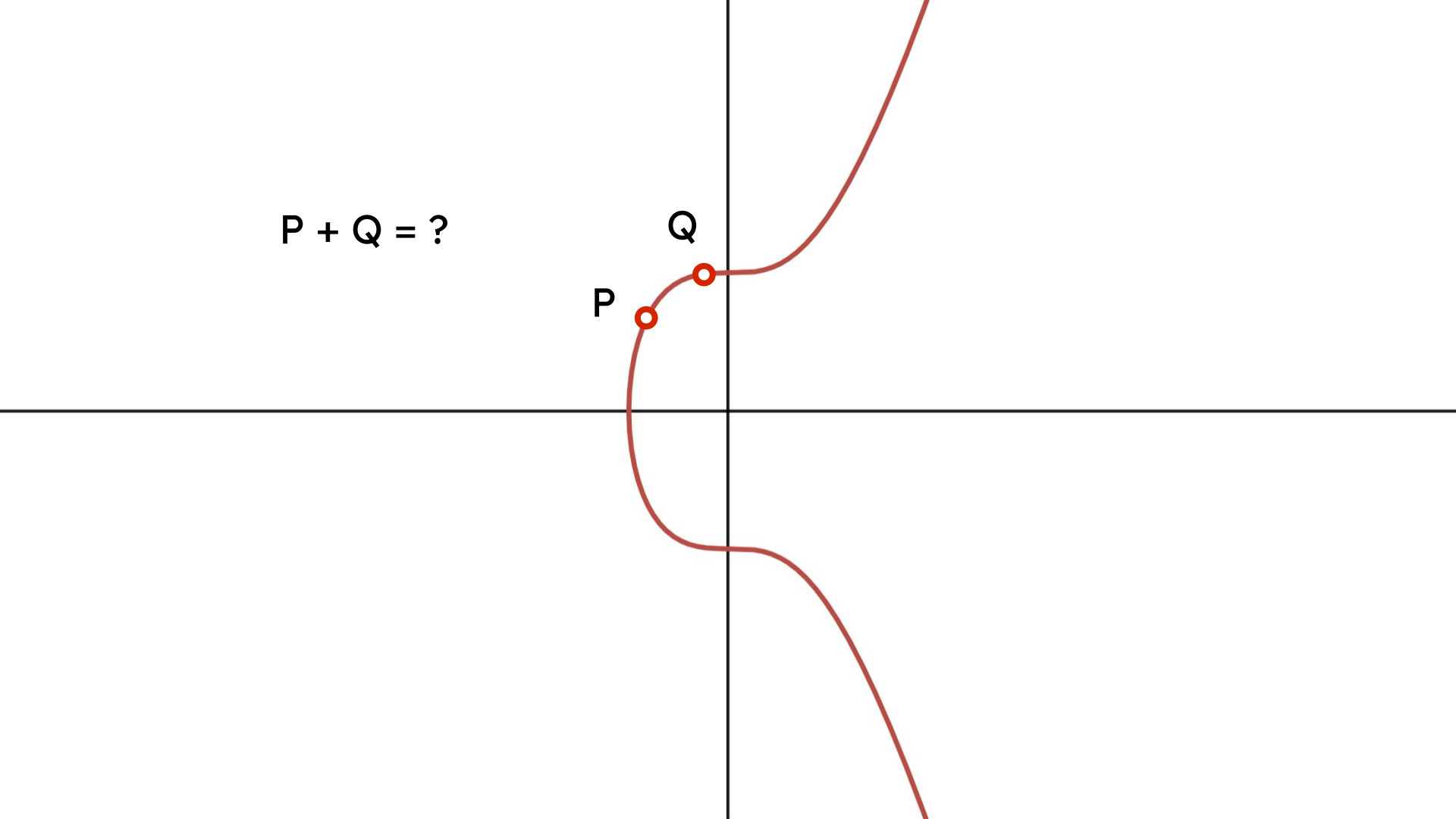




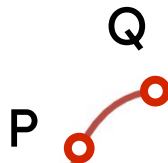
$P + Q = ?$

P

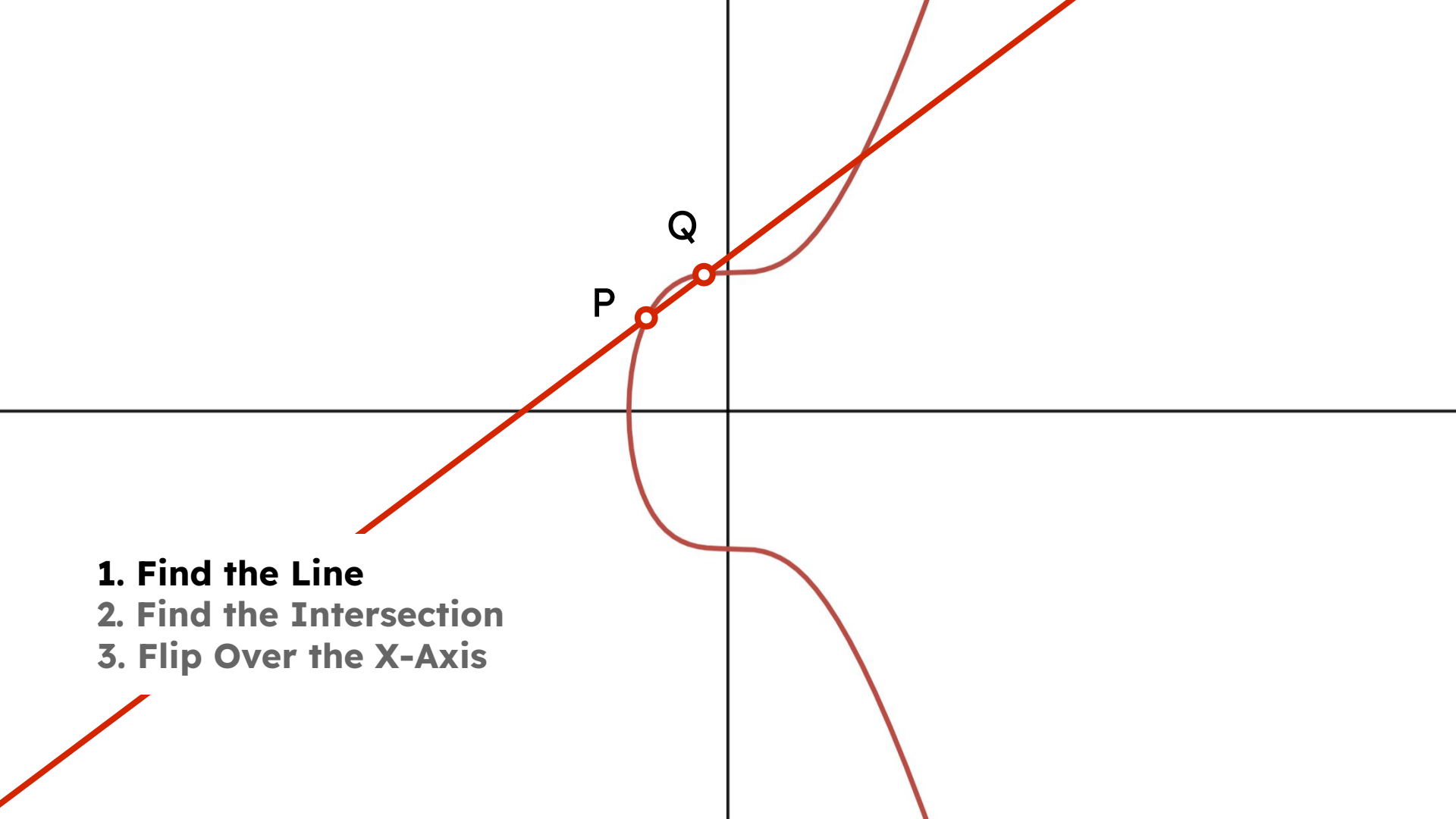
Q



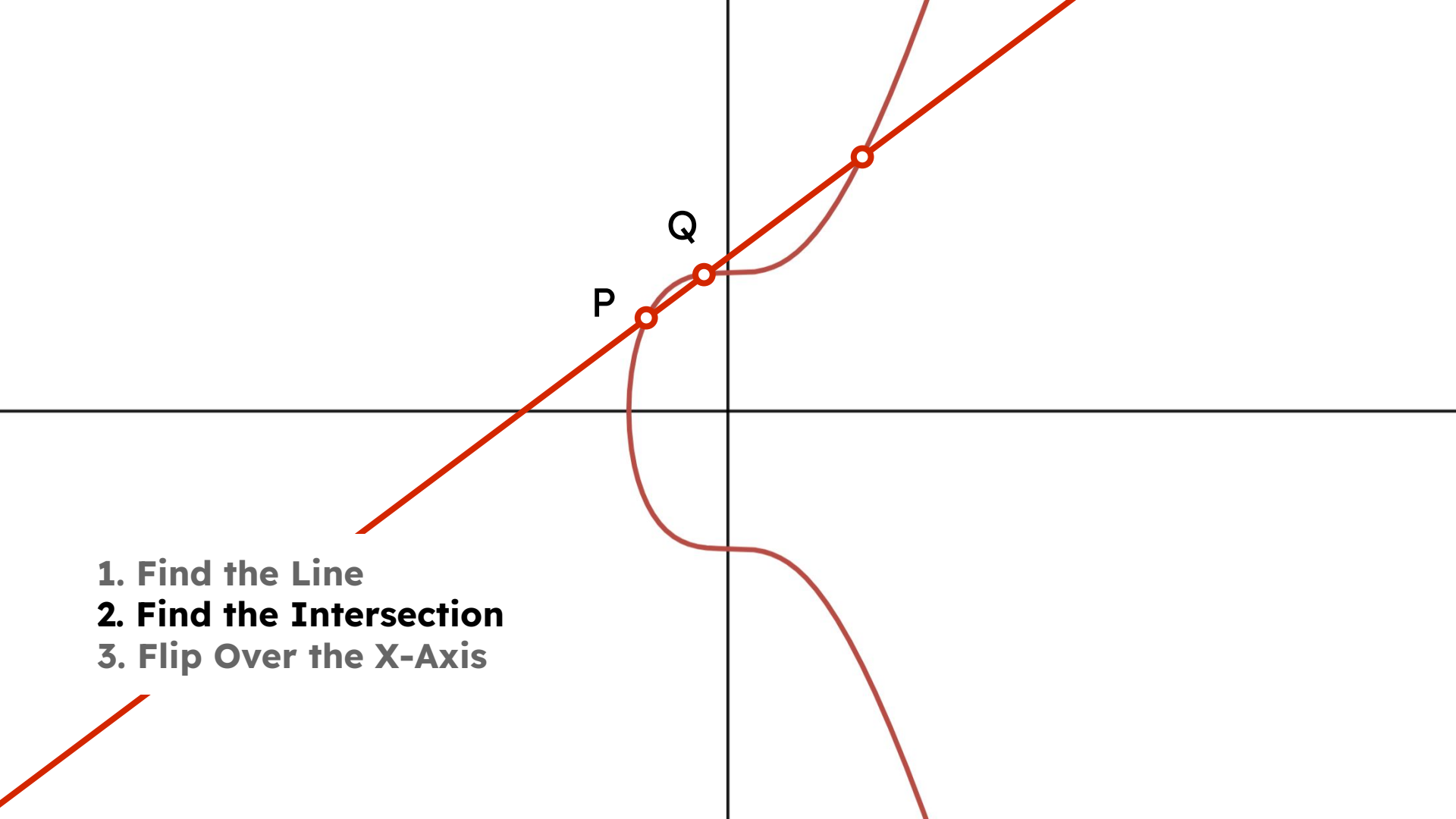
$$P + Q = ?$$



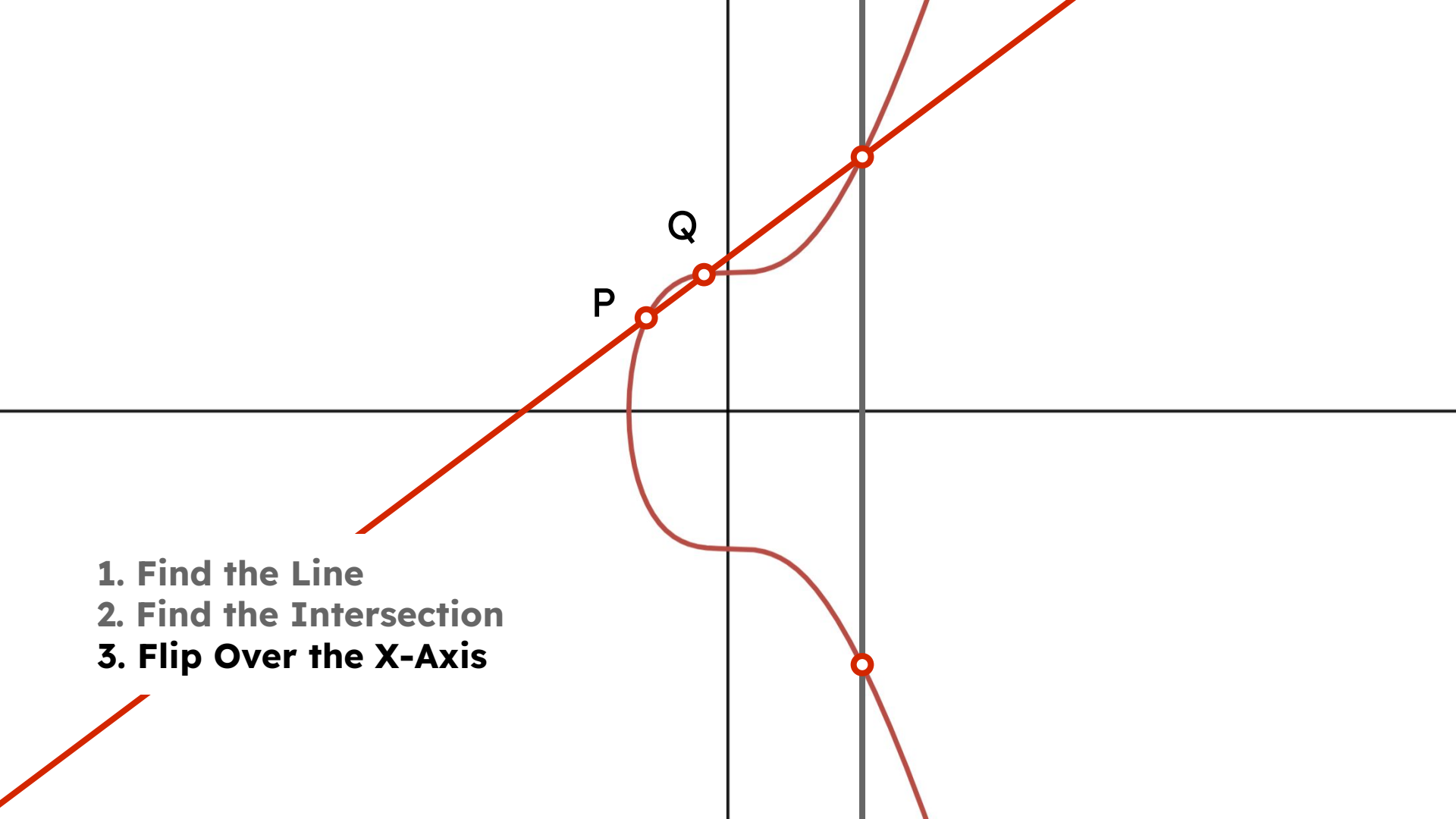
1. Find the Line
2. Find the Intersection
3. Flip Over the X-Axis



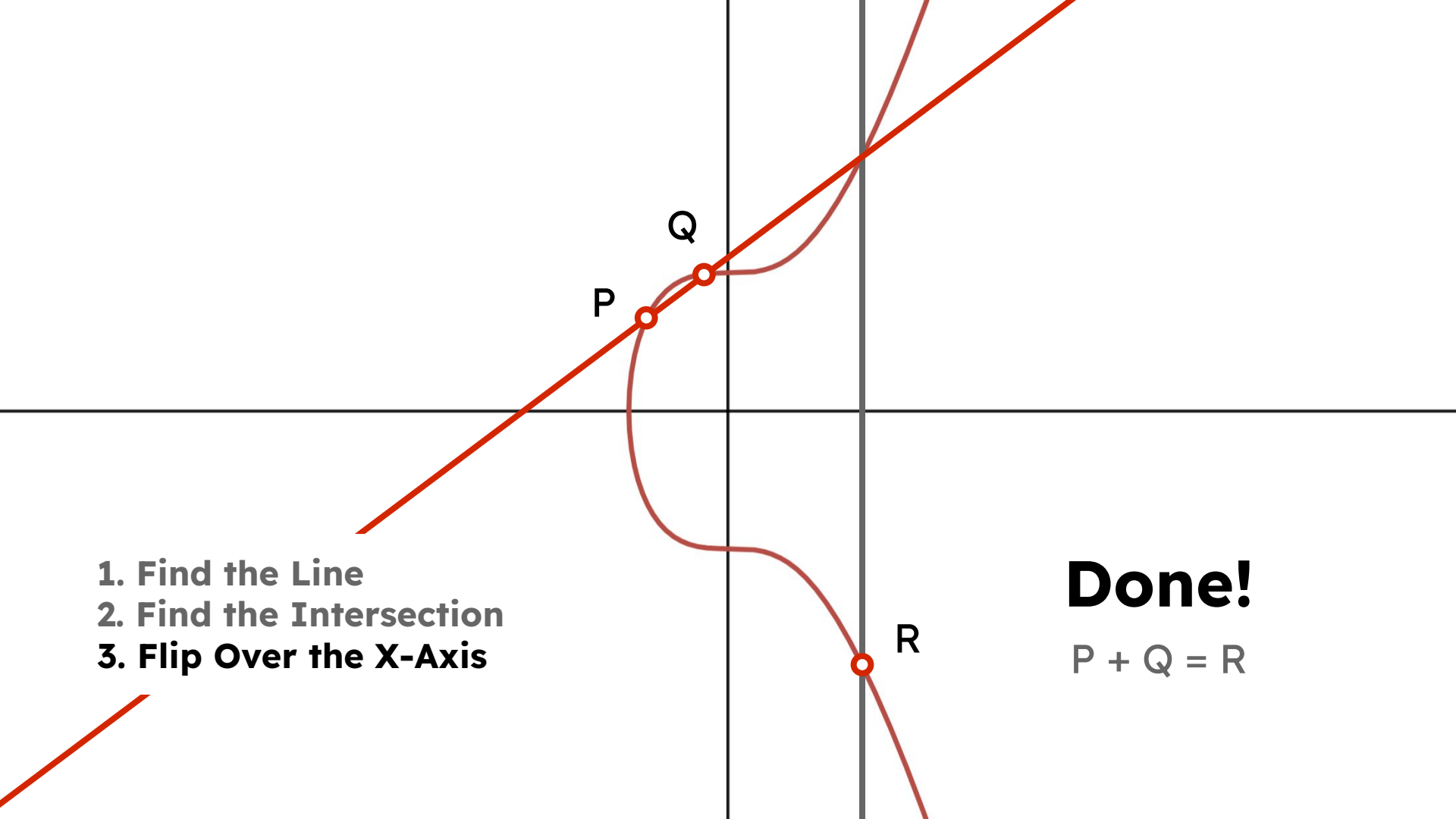
- 1. Find the Line**
- 2. Find the Intersection**
- 3. Flip Over the X-Axis**



1. Find the Line
2. Find the Intersection
3. Flip Over the X-Axis

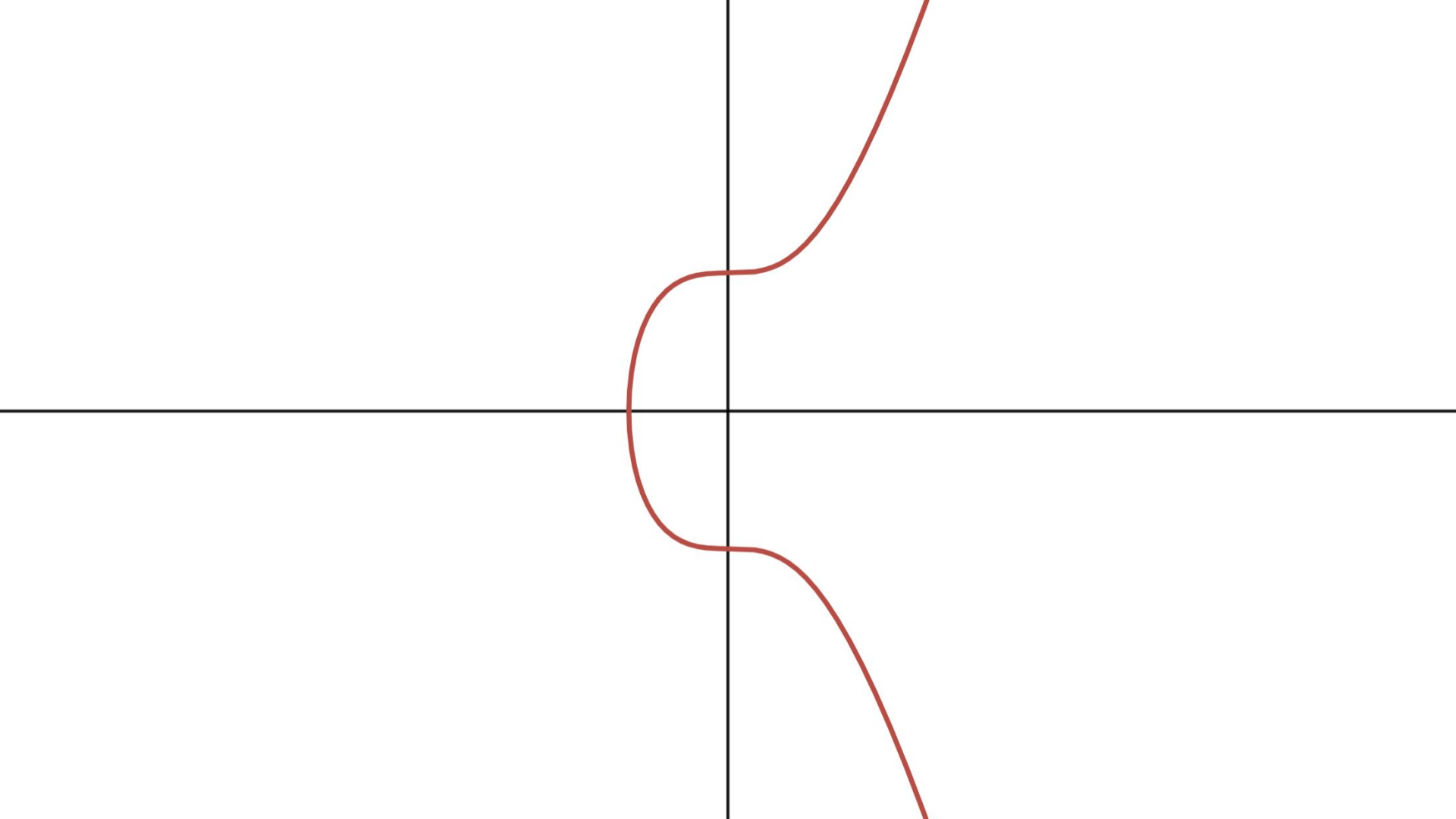


1. Find the Line
2. Find the Intersection
3. Flip Over the X-Axis

- 
1. Find the Line
  2. Find the Intersection
  3. Flip Over the X-Axis

**Done!**

$$P + Q = R$$



$$2P = P + P = ?$$

P

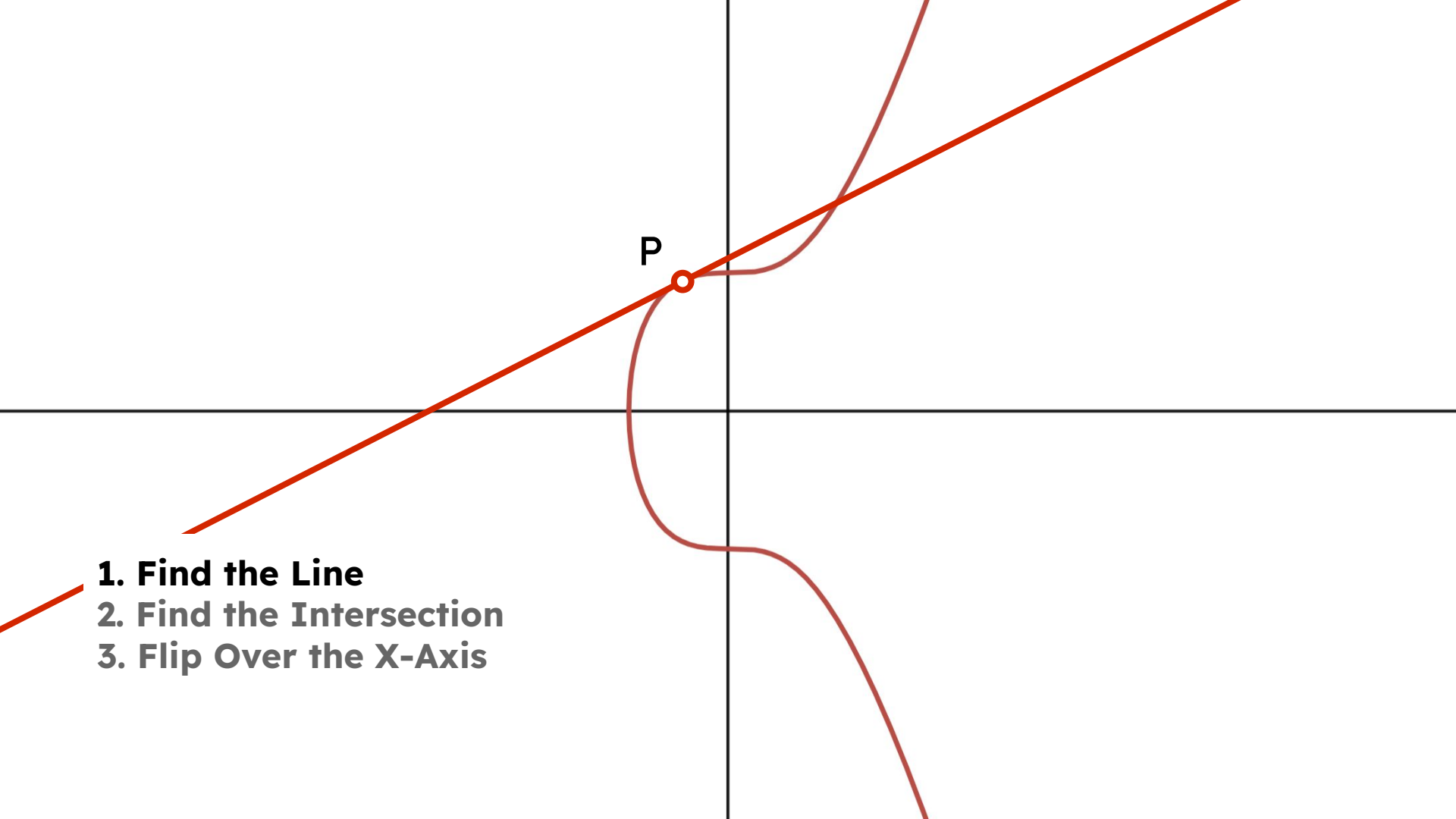


$$2P = P + P = ?$$

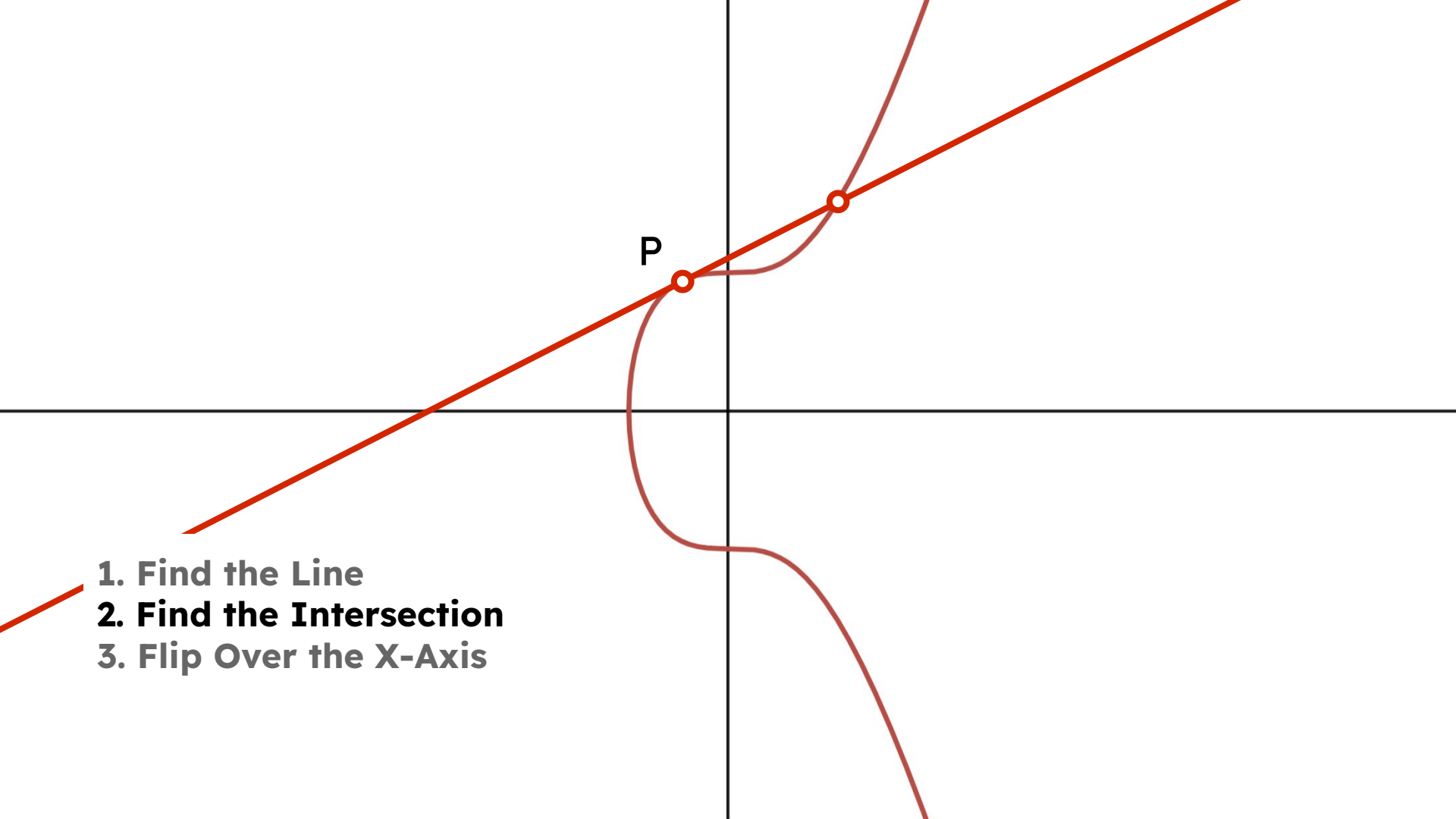
P



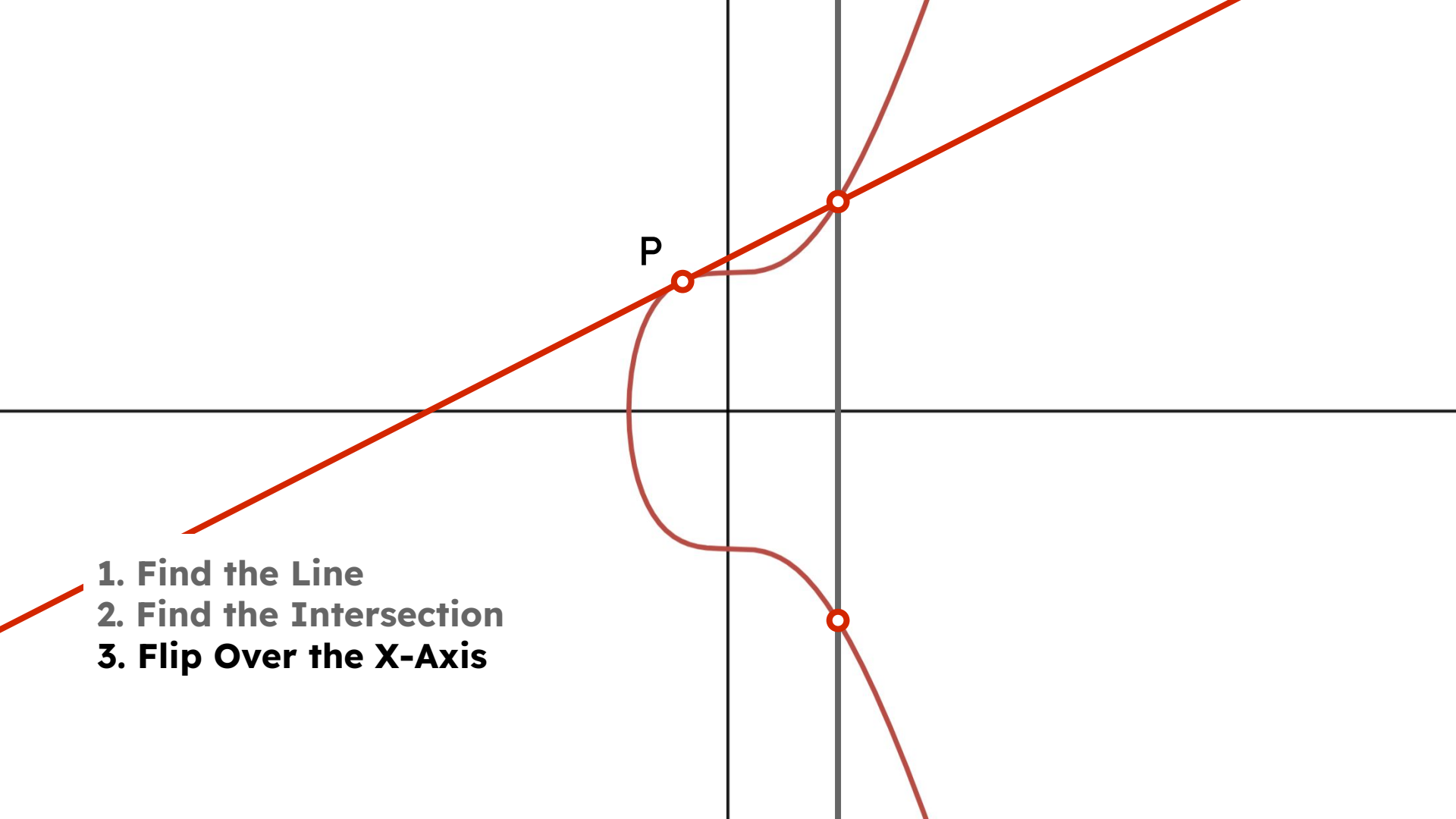
1. Find the Line
2. Find the Intersection
3. Flip Over the X-Axis

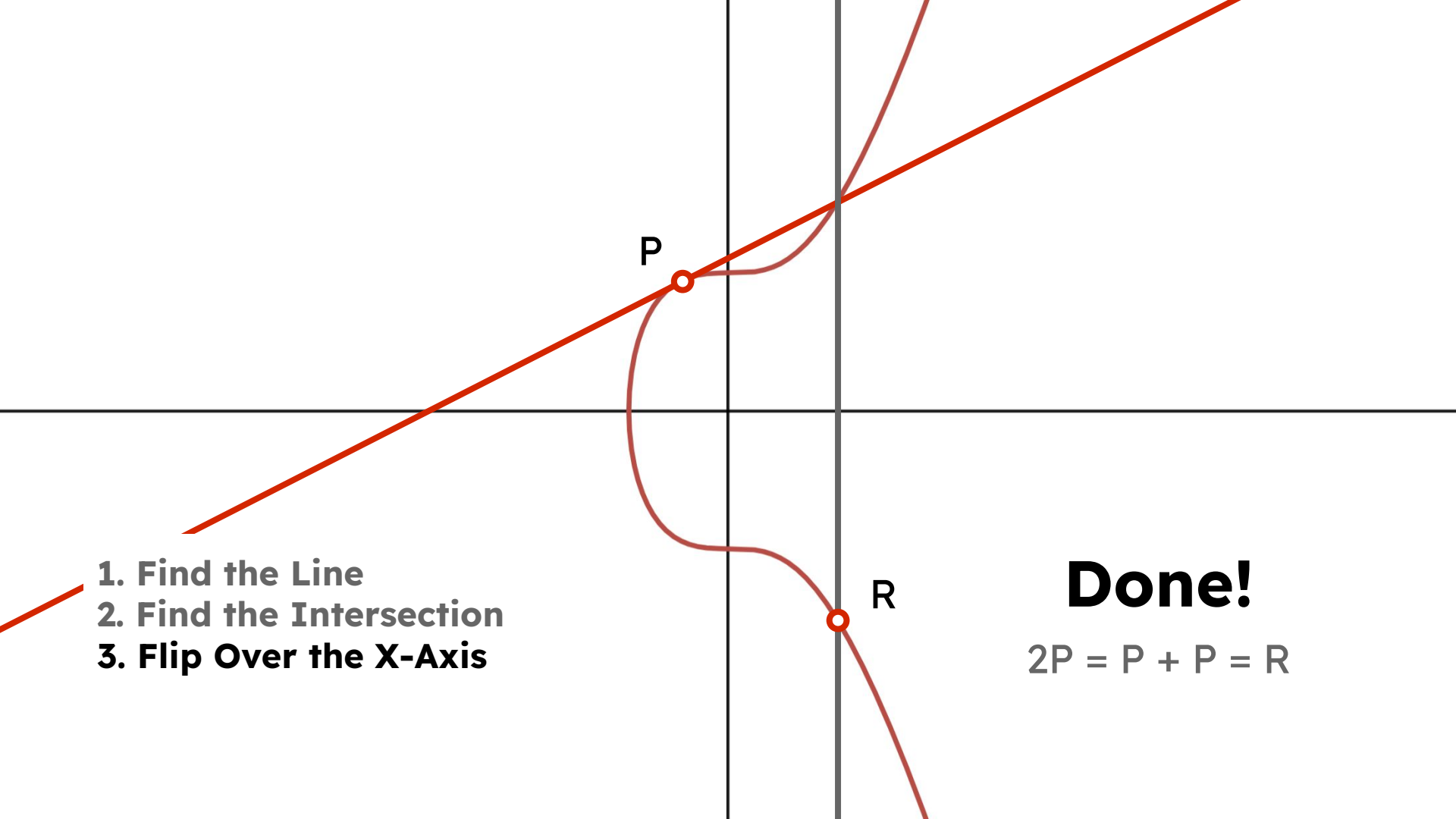


- 1. Find the Line**
- 2. Find the Intersection**
- 3. Flip Over the X-Axis**



1. Find the Line
2. Find the Intersection
3. Flip Over the X-Axis





1. Find the Line
2. Find the Intersection
3. Flip Over the X-Axis

**Done!**

$$2P = P + P = R$$

$k, Q$



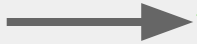
$$P = kQ$$



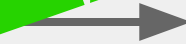
$P$

**THIS IS EASY!**

$k, Q$



$P, Q$



$P$

P, Q

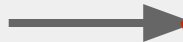


$$P = kQ$$



k

P, Q



$kQ$



k

**THIS IS HARD.**

# Blockchain History

1.3



**1991**

Blockchain invented to timestamp digital documents

**2004**

The first “proof-of-work” algorithm is formalized

**2008**

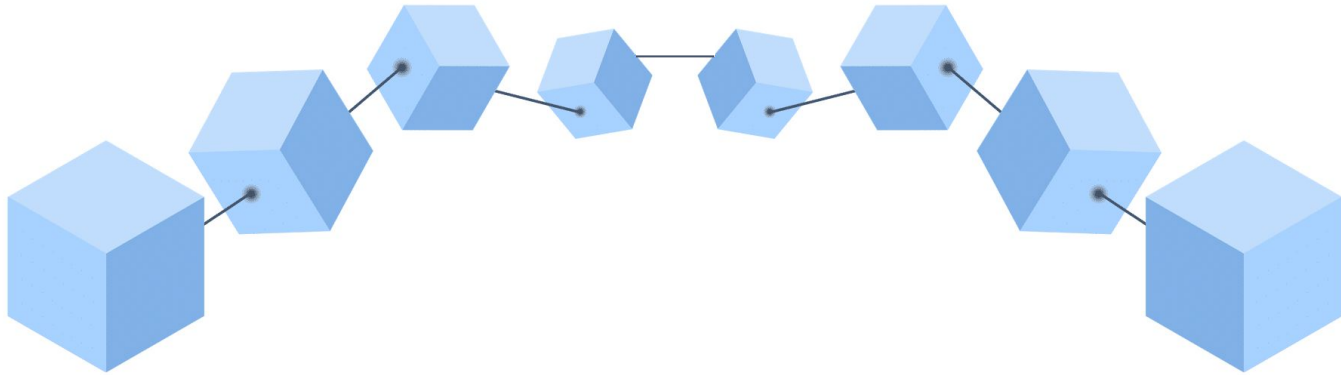
Bitcoin, the first decentralized currency, is released

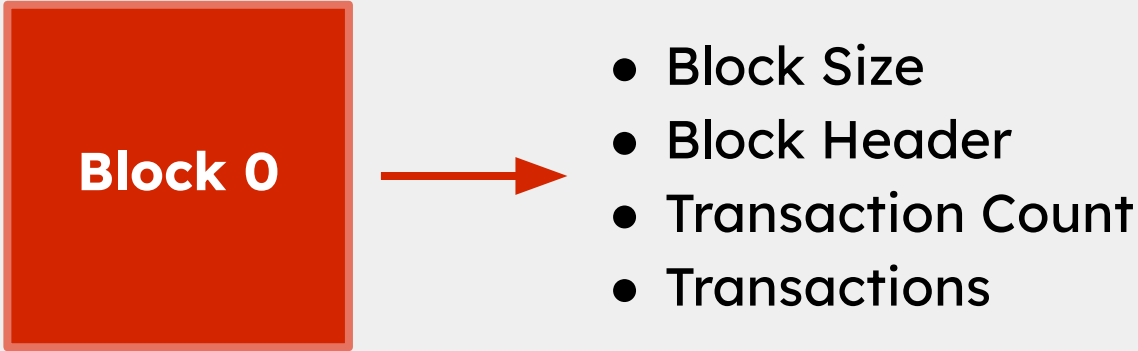
**Today**

Blockchain is widely adopted across many different fields

# The Blocks of the Chain

1.4

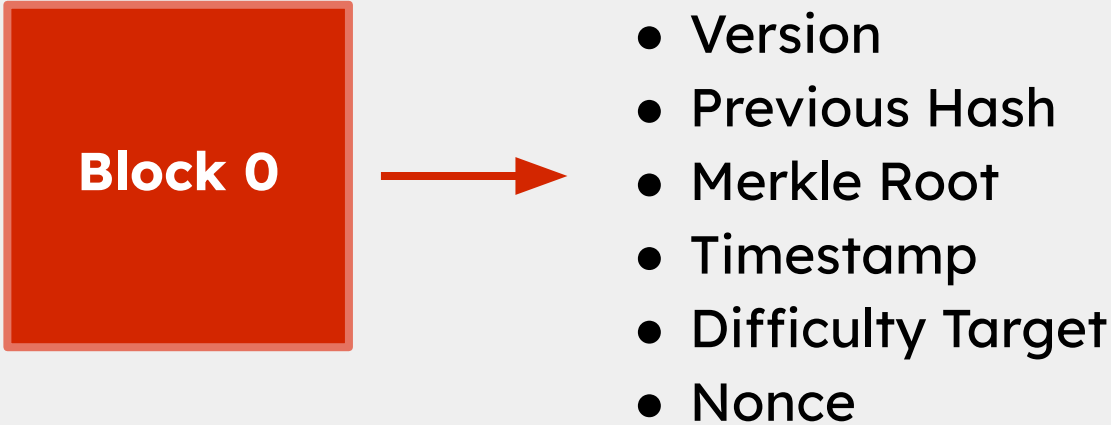




**Block 0**

A diagram illustrating the components of a block. On the left, a red square labeled "Block 0" has a red arrow pointing to a bulleted list of its components on the right.

- Block Size
- Block Header
- Transaction Count
- Transactions



**Block 0**

A diagram illustrating the structure of a block header. On the left, a red square labeled 'Block 0' has an orange arrow pointing to a list of fields under the heading 'Block Header'.

## **Block Header**

- Version
- Previous Hash
- Merkle Root
- Timestamp
- Difficulty Target
- Nonce

```
graph LR; A[Block 0] --> B[Hash Function]; B --> C[000D...AE27<br/>Block Hash];
```

**Block 0**

**Hash  
Function**

000D...AE27

Block Hash

```
graph TD; A[Block 0] --> B[Block Size]; A --> C[Transaction Count]; A --> D[Transactions];
```

**Block 0**

**Block Size**

Size of the  
block, in bytes

**Transaction Count**

Number of  
transactions

**Transactions**

Transaction data, in  
a Merkle tree

## Block 0

Hash: E7A6  
Prev: 0000

## Block 1

Hash: 89C1  
Prev: E7A6

## Block 2

Hash: 01ED  
Prev: 89C1



**Block 0**

Hash: E7A6  
Prev: 0000

**Block 1**

Hash: 89C1  
Prev: E7A6

**Block 2**

Hash: 01ED  
Prev: 89C1



Genesis Block

# Proof-of-Work Algorithm

1.5



## Block 0

Hash: E7A6  
Prev: 0000

## Block 1

Hash: 89C1  
Prev: E7A6

## Block 2

Hash: 01ED  
Prev: 89C1



## Block 0

Hash: E7A6  
Prev: 0000

## Block 1

Hash: 89C1  
Prev: E7A6

## Block 2

Hash: 01ED  
Prev: 89C1



## Block 0

Hash: AB12  
Prev: 0000

## Block 1

Hash: 89C1  
Prev: E7A6

## Block 2

Hash: 01ED  
Prev: 89C1



Changing data  
changes hash

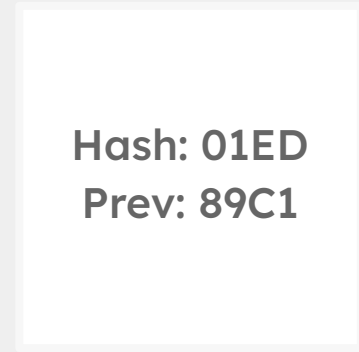
**Block 0**



**Block 1**



**Block 2**



Previous hash  
is now invalid

## Block 0

Hash: AB12  
Prev: 0000

## Block 1

Hash: 89C1  
Prev: AB12

## Block 2

Hash: 01ED  
Prev: 89C1



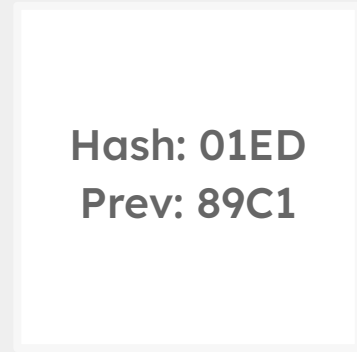
**Block 0**



**Block 1**



**Block 2**



## Block 0

Hash: AB12  
Prev: 0000

## Block 1

Hash: C789  
Prev: AB12

## Block 2

Hash: 01ED  
Prev: 89C1



**Block 0**

Hash: AB12  
Prev: 0000

**Block 1**

Hash: C789  
Prev: AB12

**Block 2**

Hash: 01ED  
Prev: 89C1



**Block 0**

Hash: AB12  
Prev: 0000

**Block 1**

Hash: C789  
Prev: AB12

**Block 2**

Hash: 01ED  
Prev: C789



**Block 0**

Hash: AB12  
Prev: 0000

**Block 1**

Hash: C789  
Prev: AB12

**Block 2**

Hash: E14D  
Prev: C789



**Block 0**

Hash: AB12  
Prev: 0000

**Block 1**

Hash: C789  
Prev: AB12

**Block 2**

Hash: E14D  
Prev: C789



Block 0

Hash: AB12  
Prev: 0000

Block 1

Hash: C789  
Prev: AB12

Block 2

Hash: E14D  
Prev: C789

**THIS IS EASY!**

## Block 0

Nonce: D7  
Hash: 00AE  
Prev: 0000



**New Requirement:**

Hash must be less than 00FF

## Block 0

**Nonce: D7**  
Hash: 00AE  
Prev: 0000



Nonce allows us to manipulate the hash without changing the data

## Block 0

Nonce: D7  
Hash: 00AE  
Prev: 0000

## Block 1

Nonce: 00  
Hash: 279B  
Prev: 00AE



New blocks cannot  
be added until they  
have a valid hash

## Block 0

Nonce: D7  
Hash: 00AE  
Prev: 0000

## Block 1

Nonce: 01  
Hash: 87DF  
Prev: 00AE



Changing nonce  
changes hash

## Block 0

Nonce: D7  
Hash: 00AE  
Prev: 0000

## Block 1

Nonce: 02  
Hash: 41C2  
Prev: 00AE



## Block 0

Nonce: D7  
Hash: 00AE  
Prev: 0000

## Block 1

Nonce: 03  
Hash: 40E6  
Prev: 00AE



## Block 0

Nonce: D7  
Hash: 00AE  
Prev: 0000

## Block 1

Nonce: 04  
Hash: 0076  
Prev: 00AE



Eventually, a valid  
hash is found...

## Block 0

Nonce: D7  
Hash: 00AE  
Prev: 0000

## Block 1

Nonce: 04  
Hash: 0076  
Prev: 00AE



...and the block is  
added to the chain

## Block 0

Nonce: D7  
Hash: 00AE  
Prev: 0000

## Block 1

Nonce: 04  
Hash: 0076  
Prev: 00AE

## Block 2

Nonce: 62  
Hash: 00C1  
Prev: 0076



## Block 0

Nonce: D7  
Hash: 00AE  
Prev: 0000

## Block 1

Nonce: 04  
Hash: 0076  
Prev: 00AE

## Block 2

Nonce: 62  
Hash: 00C1  
Prev: 0076



Block 0

Nonce: D7  
Hash: 00AE  
Prev: 0000

Block 1

Nonce: 04  
Hash: 0076  
Prev: 00AE

Block 2

Nonce: 62  
Hash: 00C1  
Prev: 0076

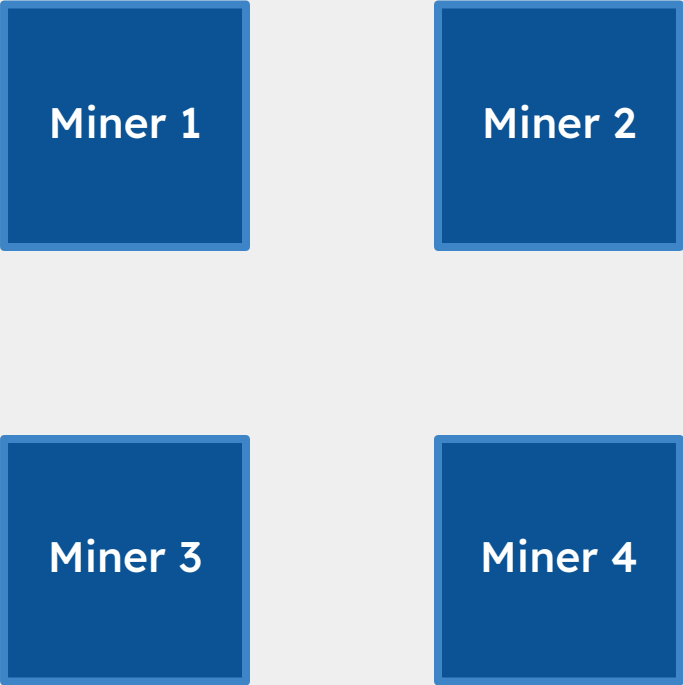
**THIS IS HARD.**

- Finding a valid hash takes 10 minutes on the Bitcoin network
- The value each hash must be less than is determined by the difficulty target
- Difficulty target updated every 2 weeks to account for network fluctuations
- Manipulating blocks earlier in the chain is prohibitively expensive

# Distribution and Consensus

1.6





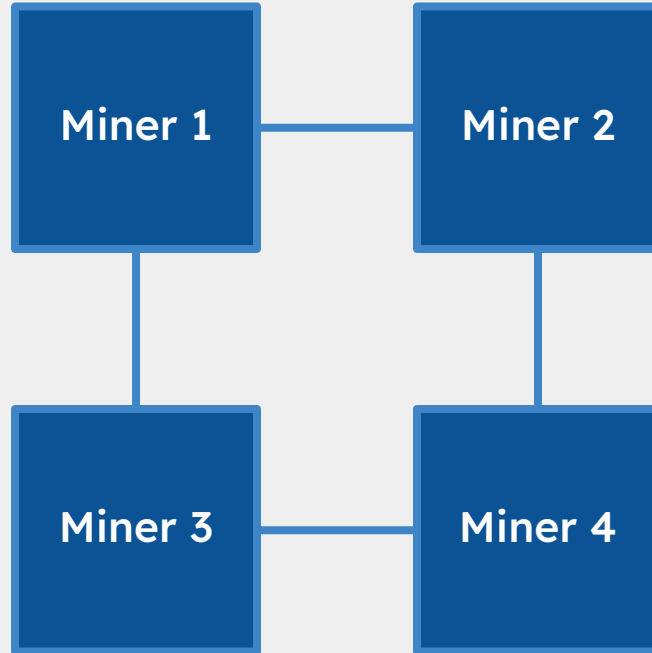
A diagram showing four miners arranged in a 2x2 grid. Each miner is represented by a blue square with a white border. The squares are labeled 'Miner 1', 'Miner 2', 'Miner 3', and 'Miner 4' in white text. The miners are arranged in two rows and two columns.

**Miner 1**

**Miner 2**

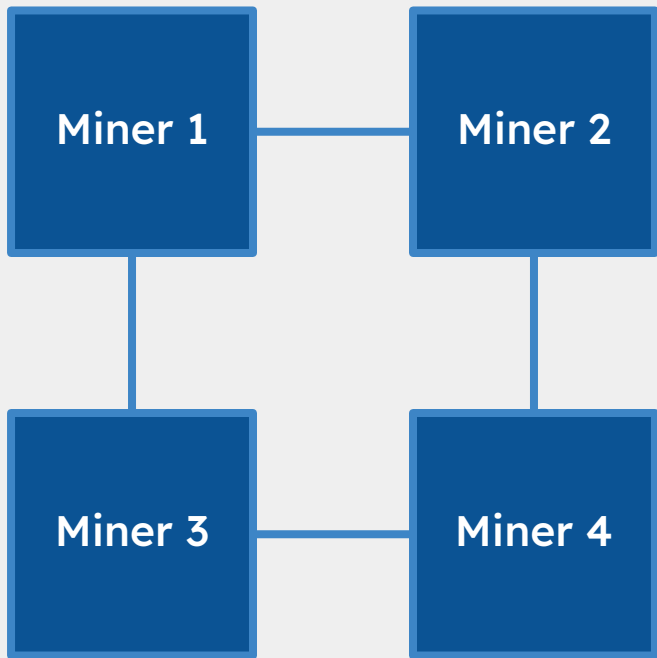
**Miner 3**

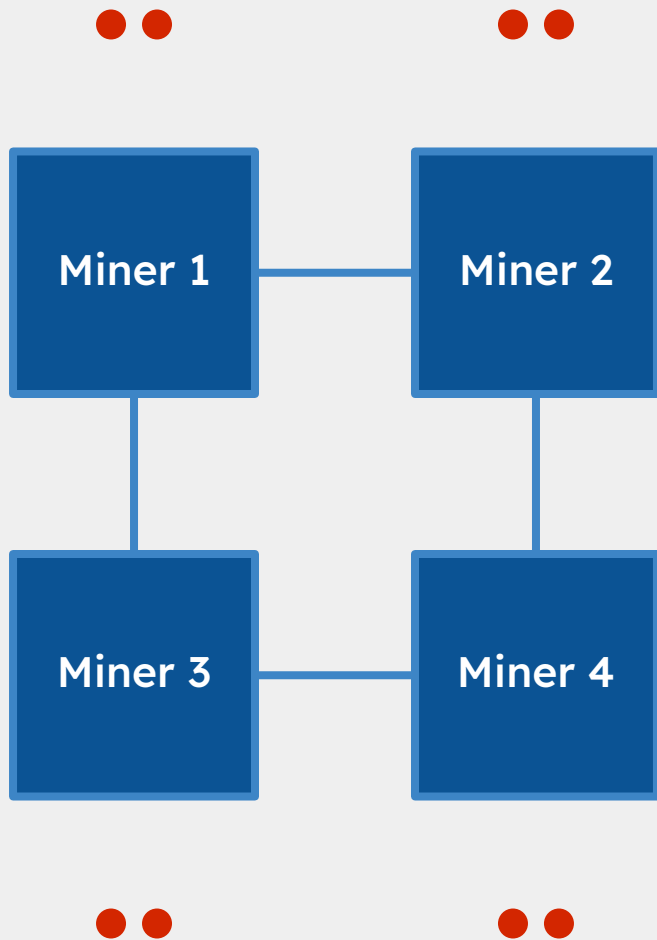
**Miner 4**



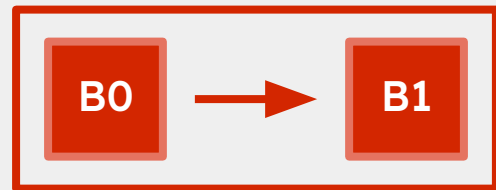


Miner 2's local blockchain

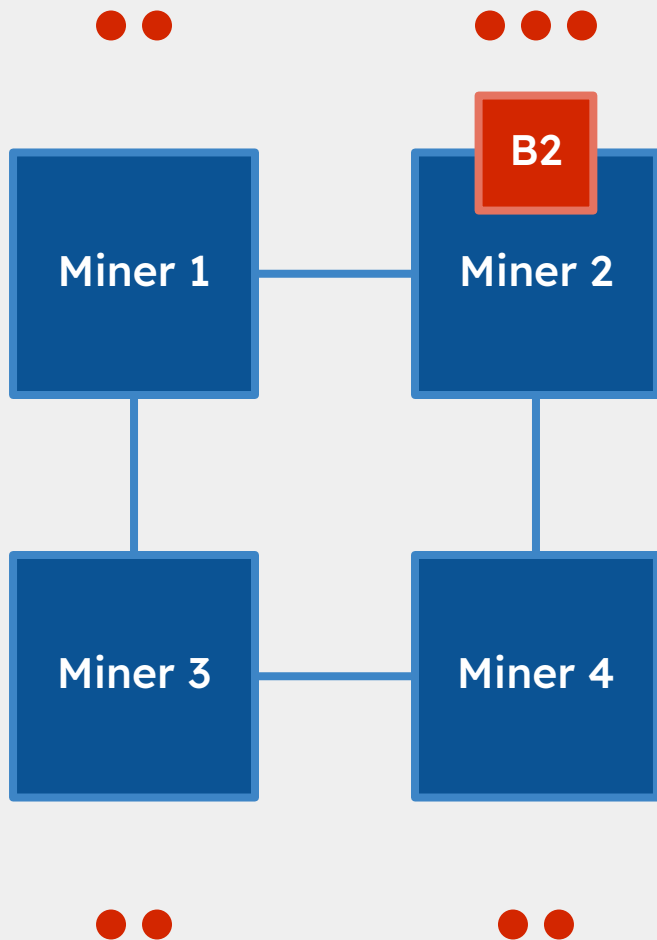




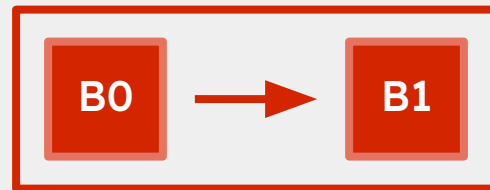
**Synced Blockchain**

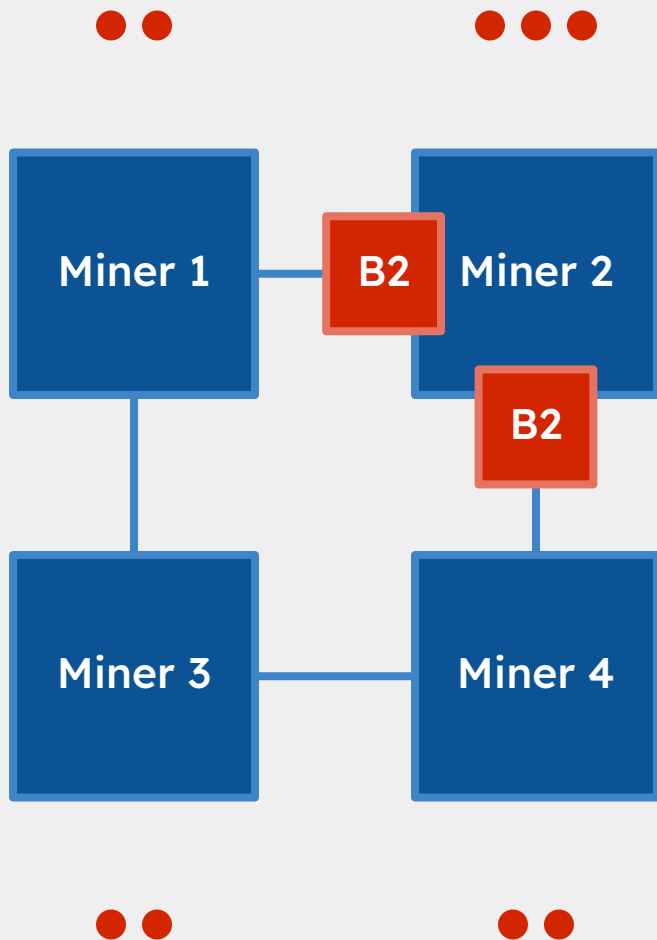


Only theoretical

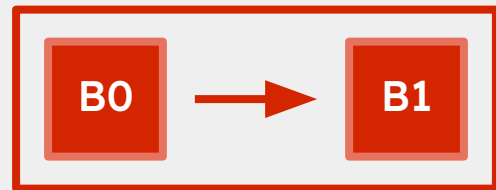


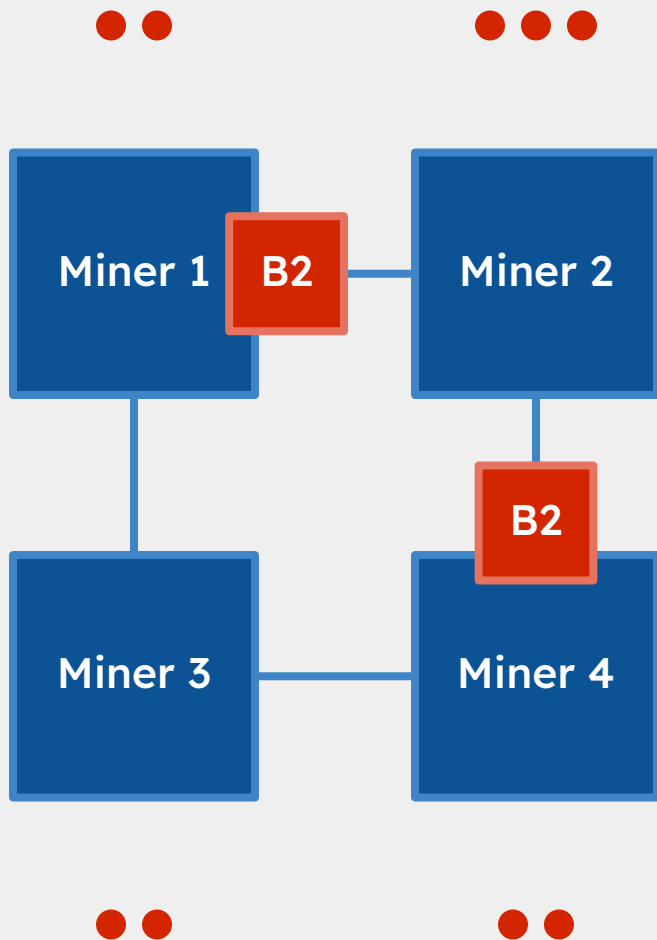
## Synced Blockchain



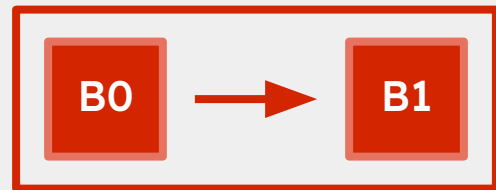


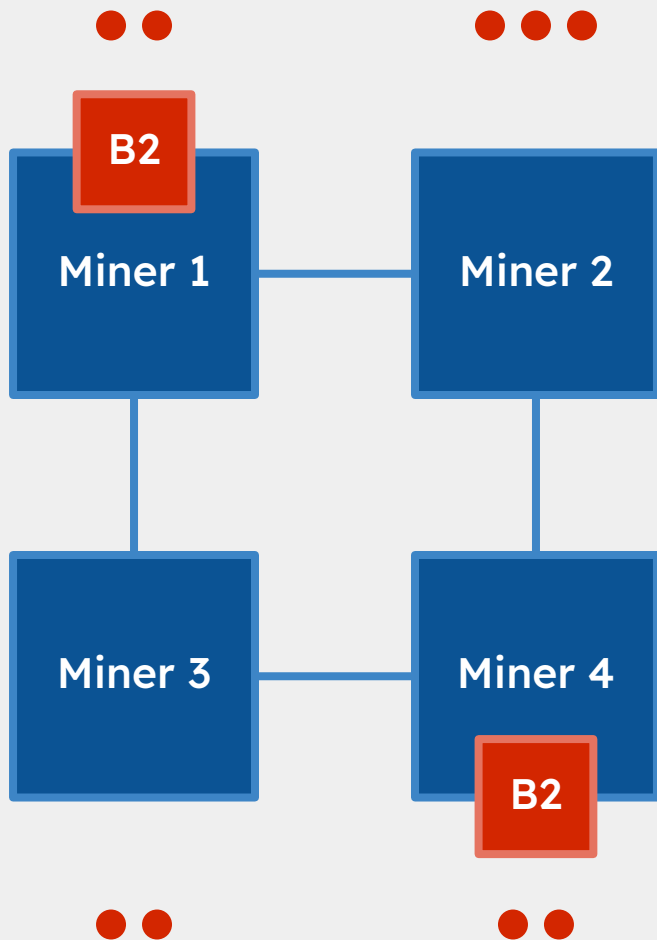
## Synced Blockchain



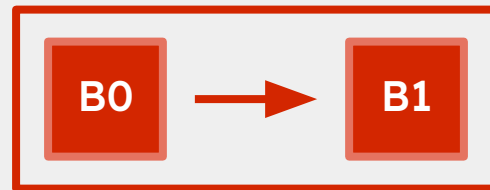


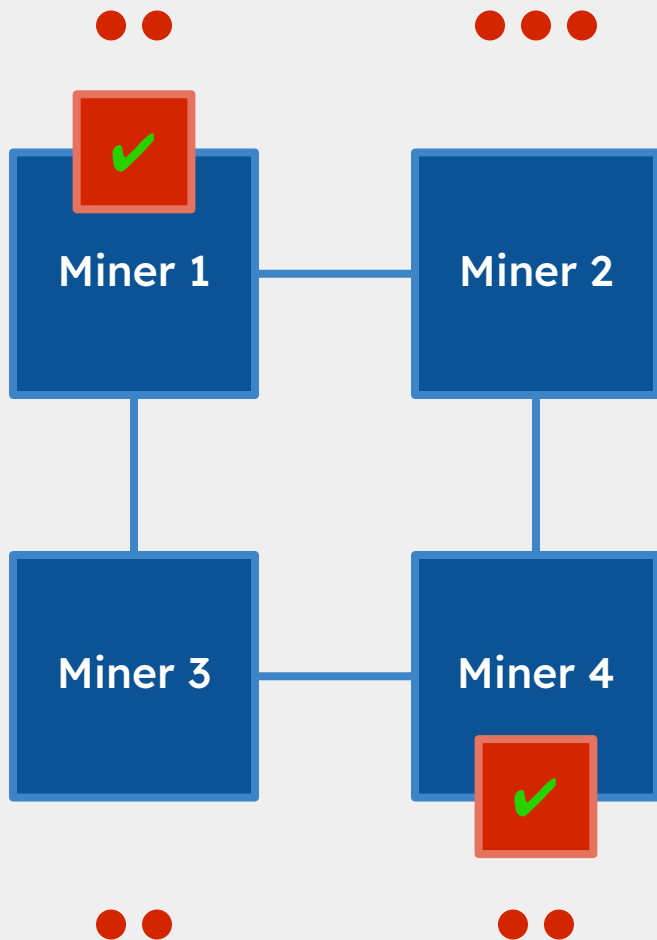
## Synced Blockchain



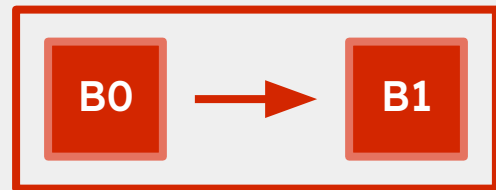


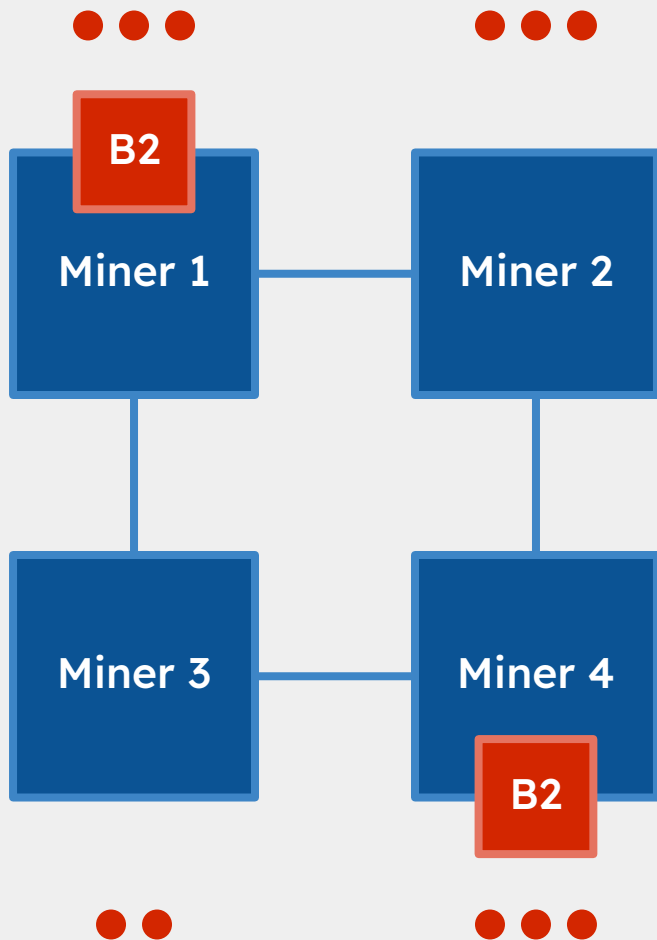
## Synced Blockchain



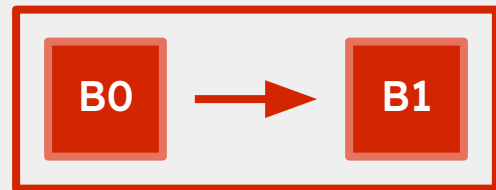


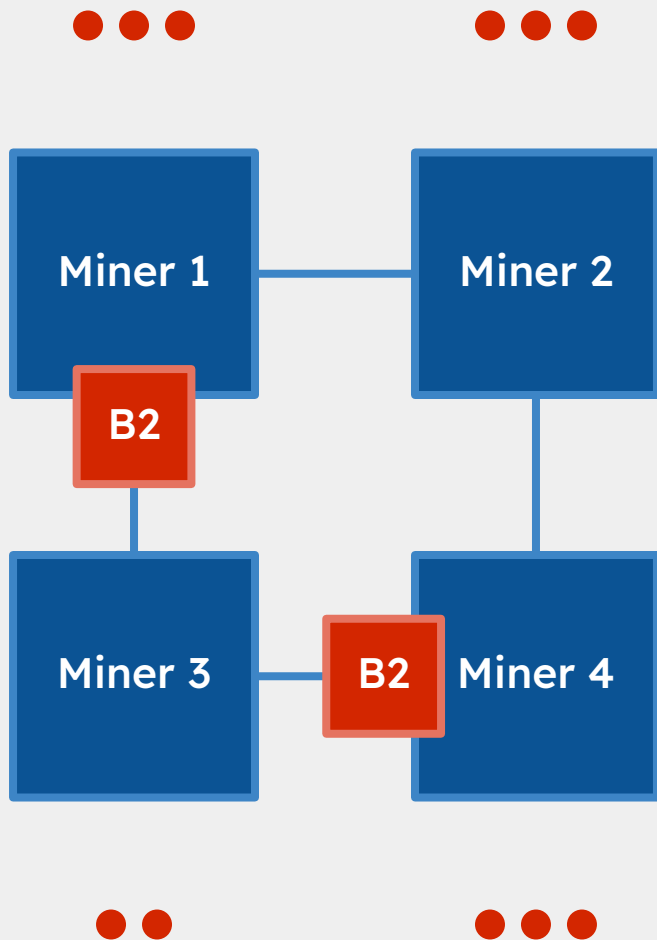
## Synced Blockchain



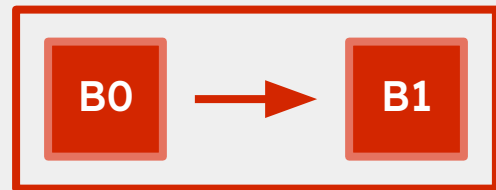


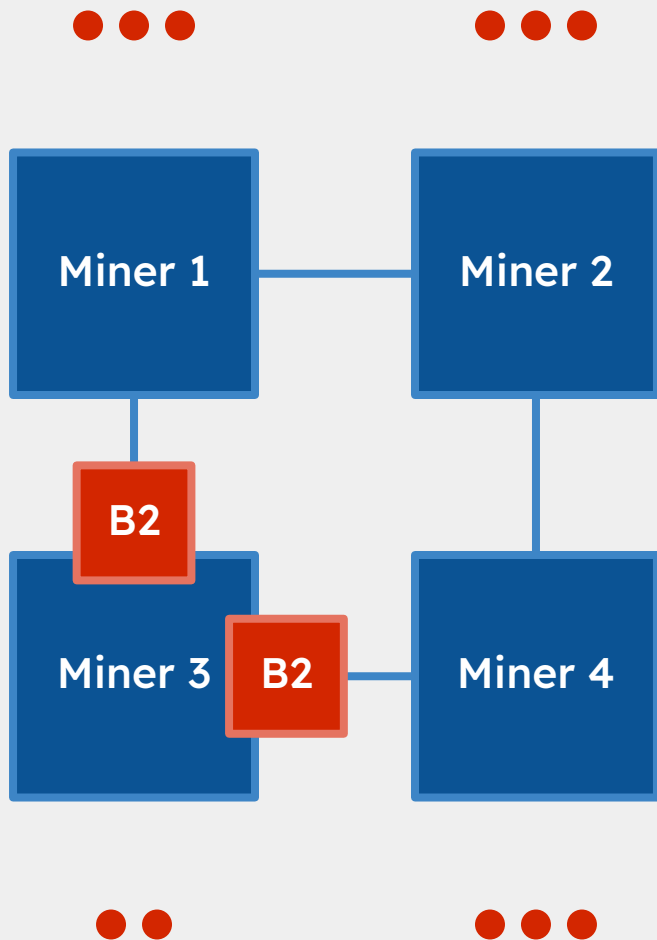
## Synced Blockchain



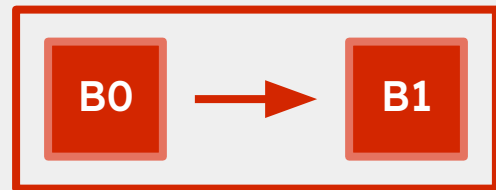


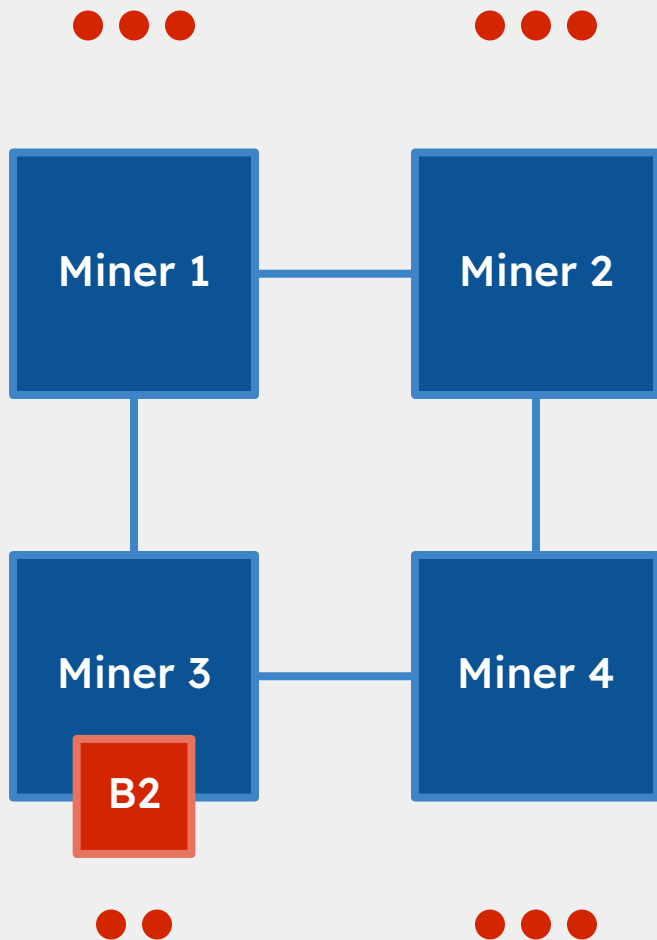
## Synced Blockchain



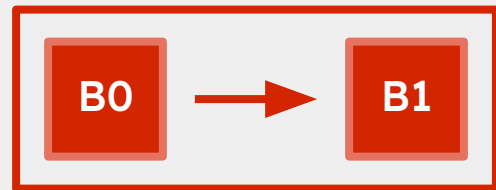


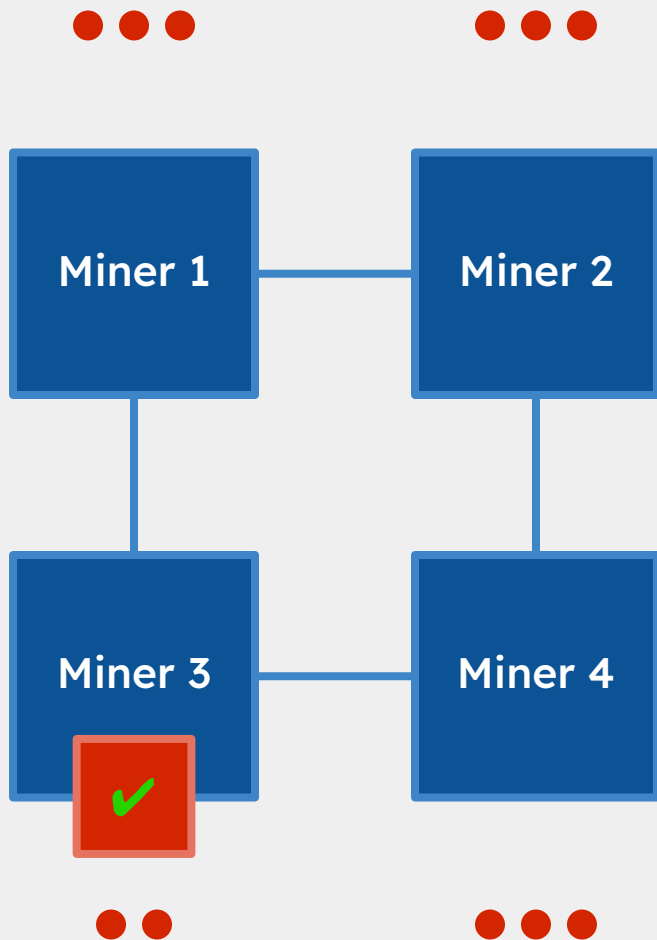
## Synced Blockchain



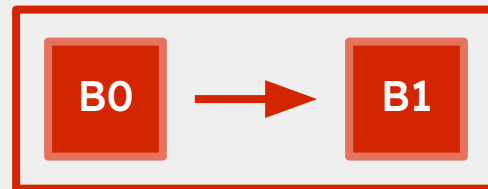


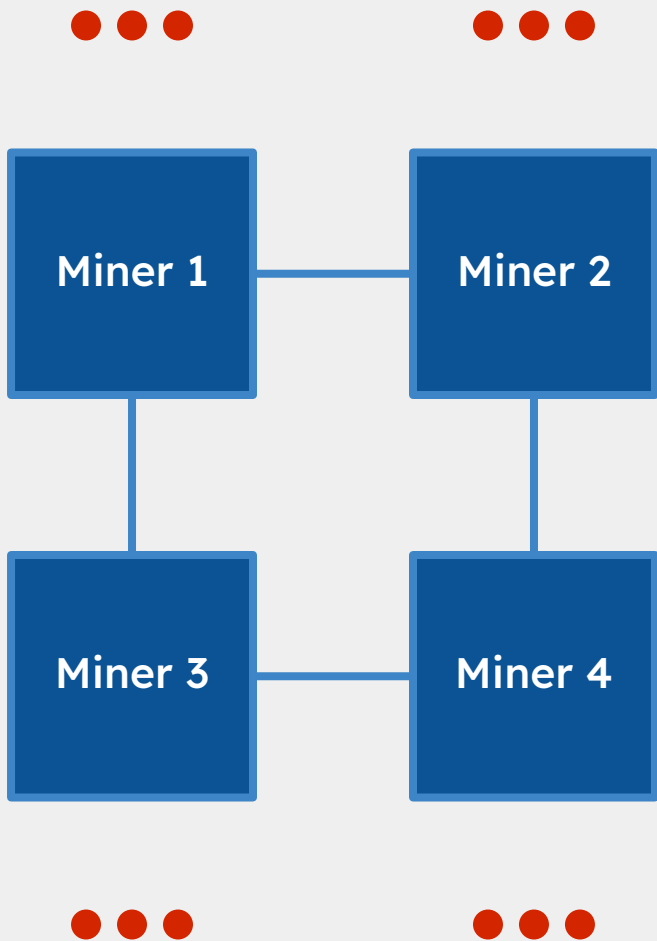
## Synced Blockchain





## Synced Blockchain





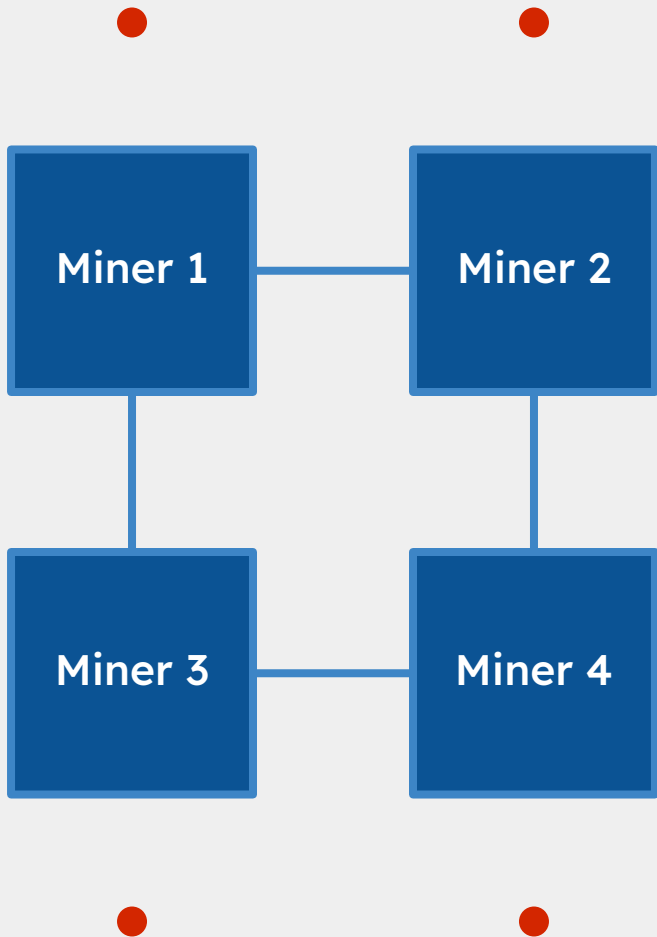
### Synced Blockchain





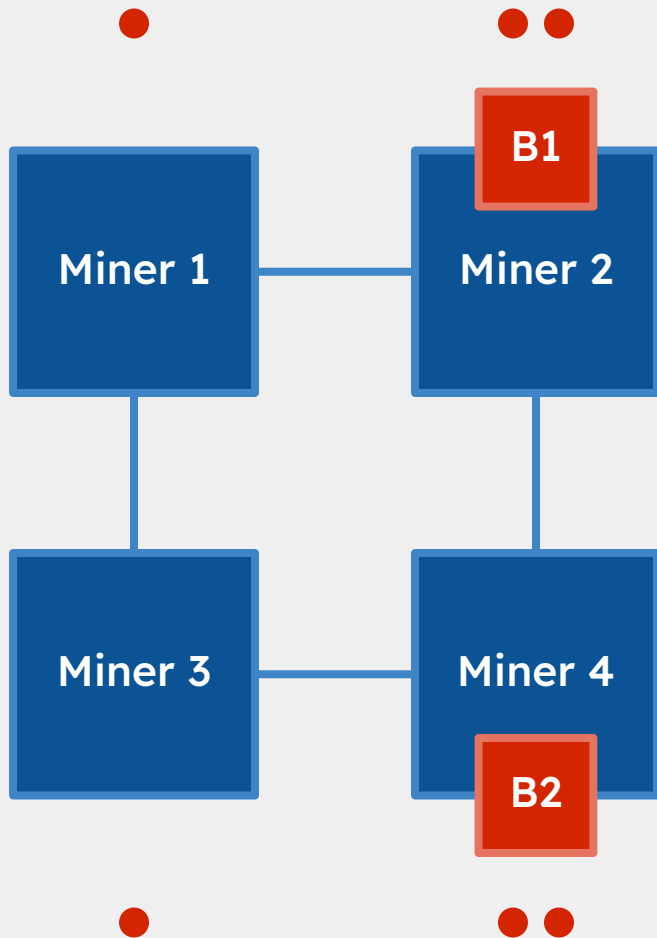
**ALL IN SYNC!**





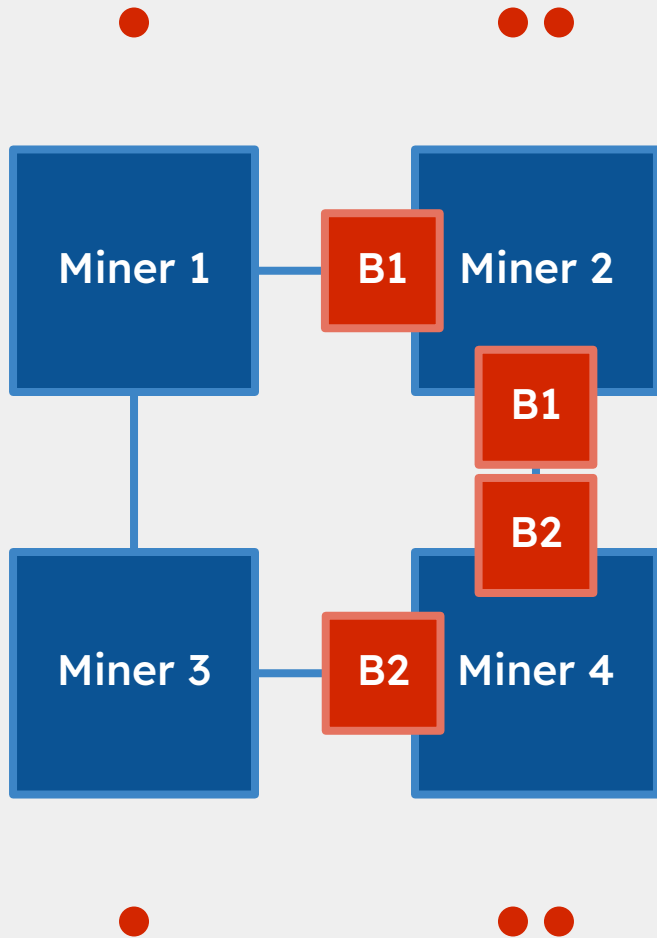
**Synced Blockchain**





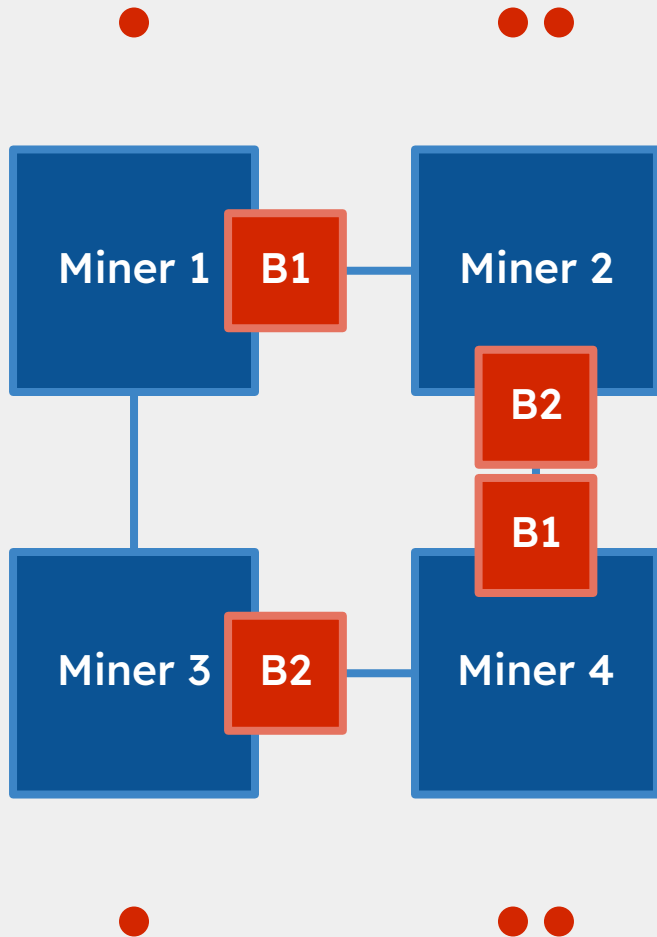
**Synced Blockchain**





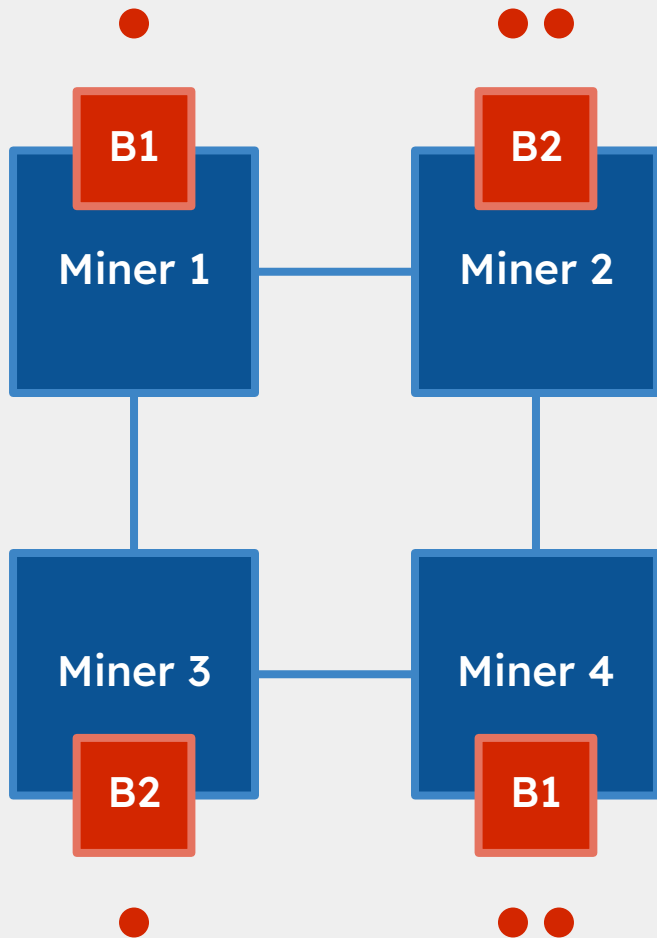
**Synced Blockchain**





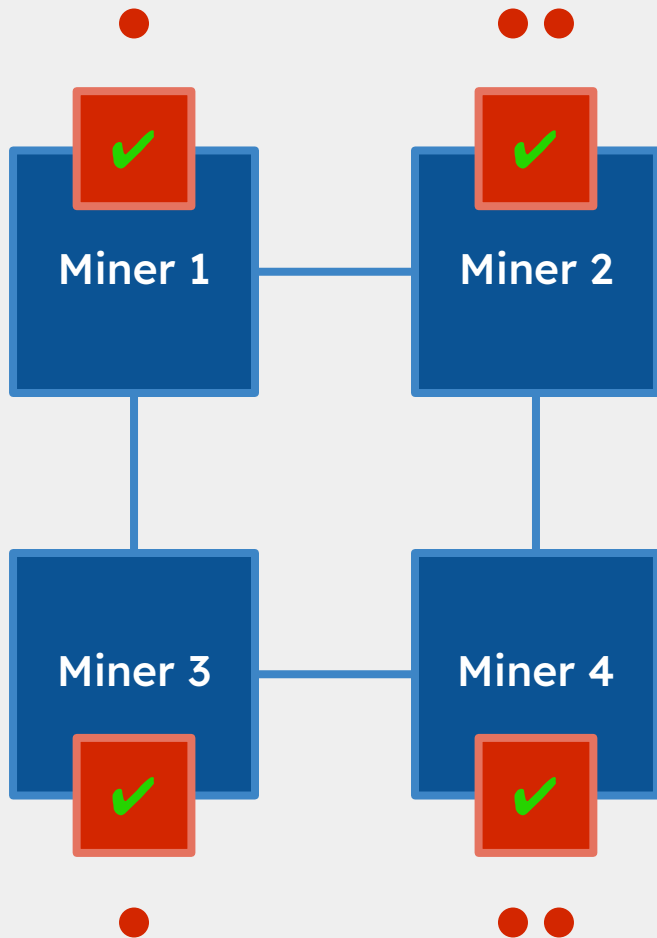
**Synced Blockchain**





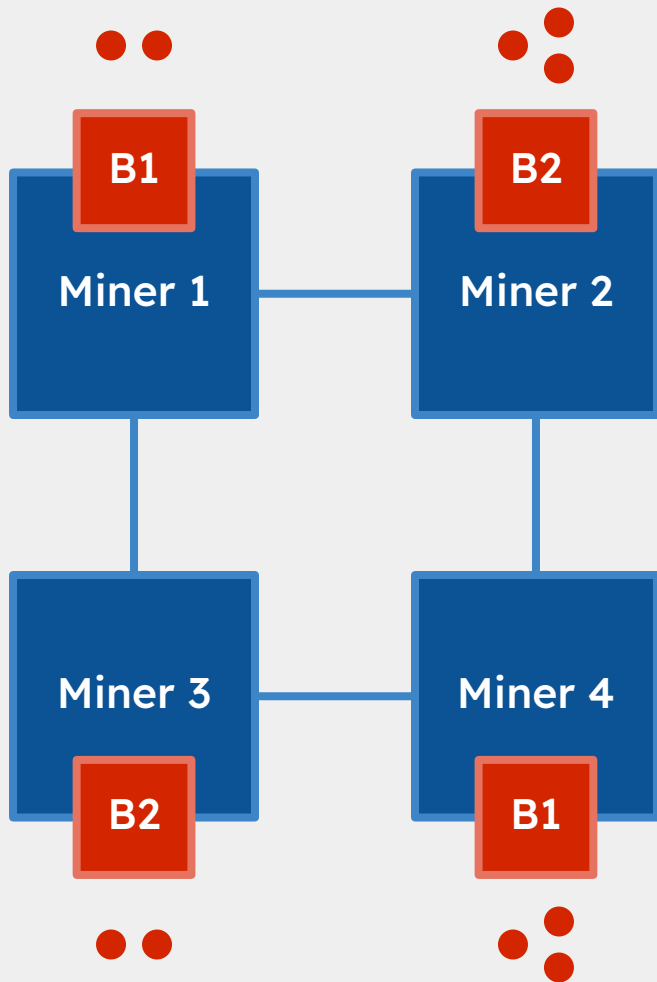
**Synced Blockchain**





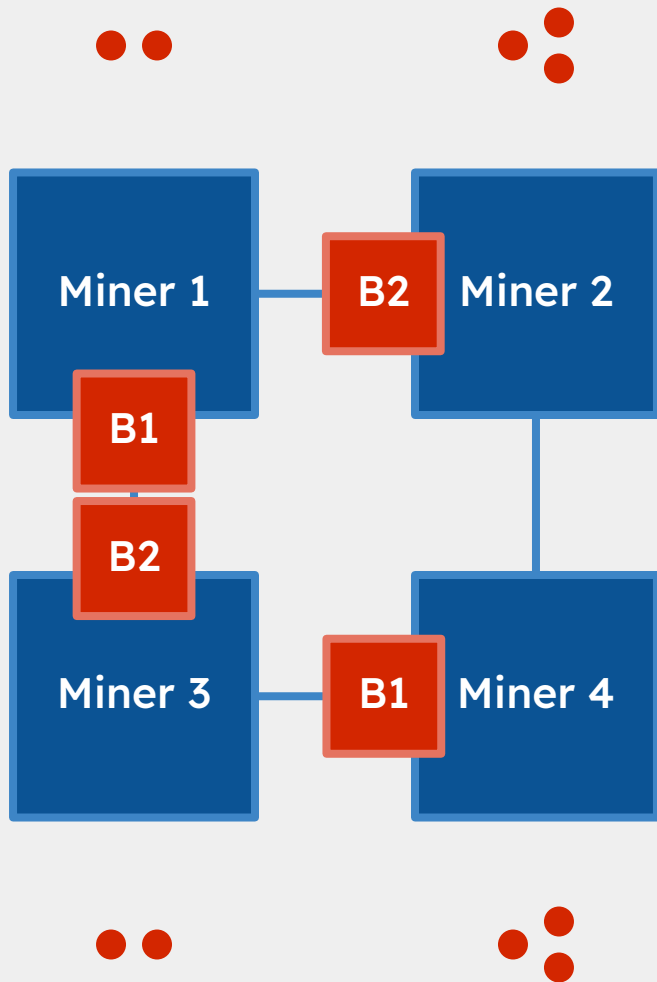
**Synced Blockchain**





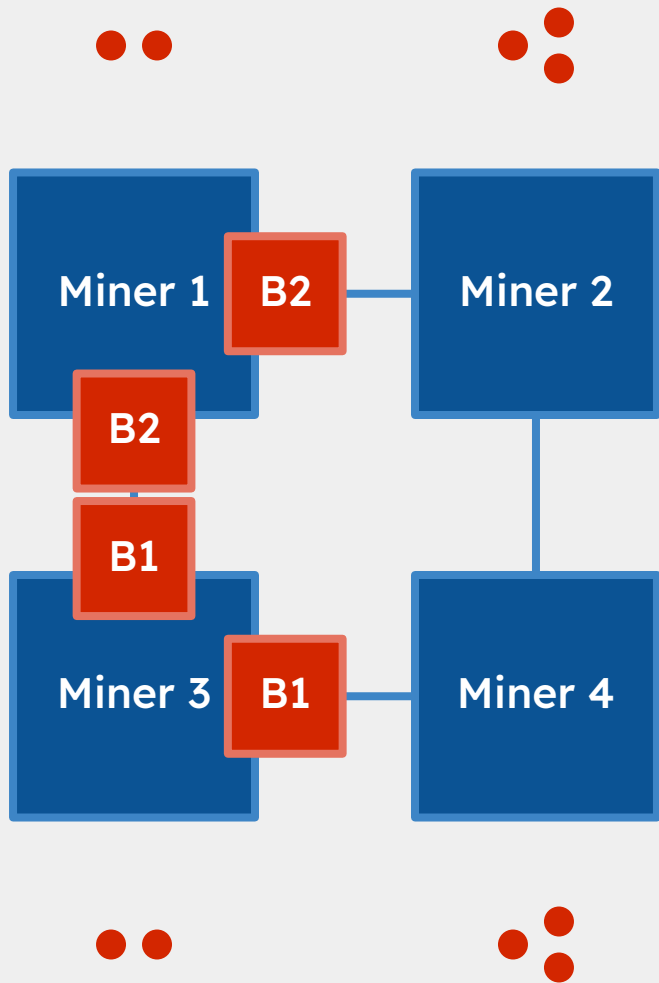
**Synced Blockchain**





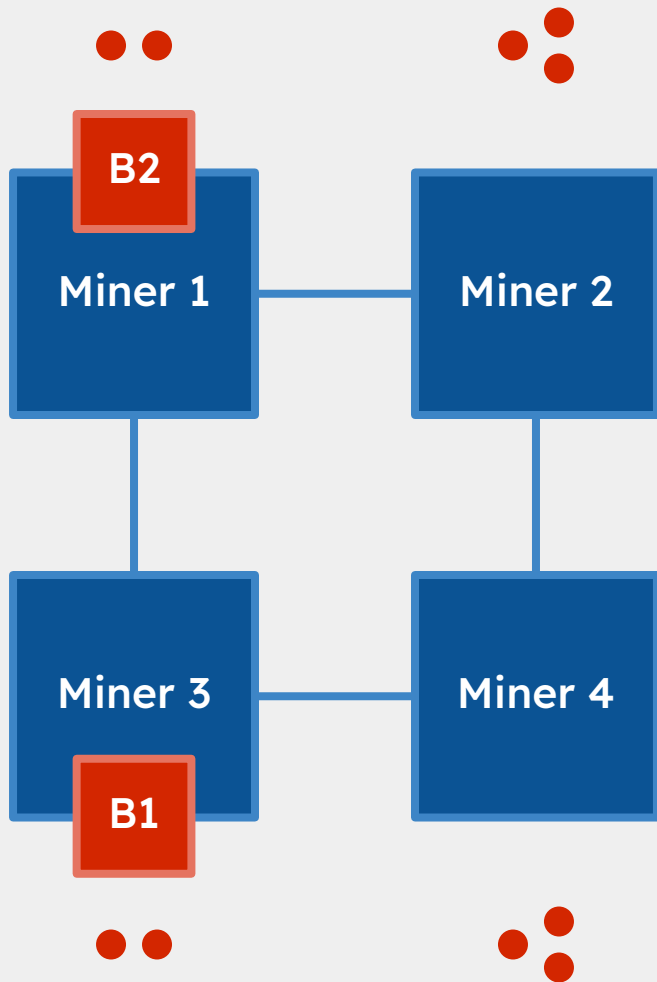
**Synced Blockchain**





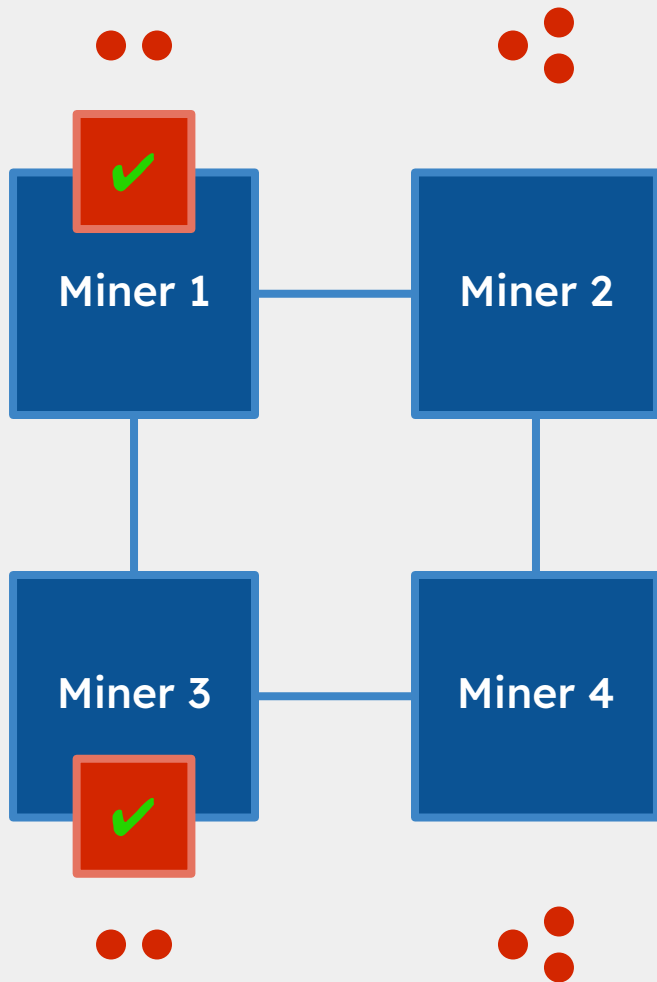
**Synced Blockchain**





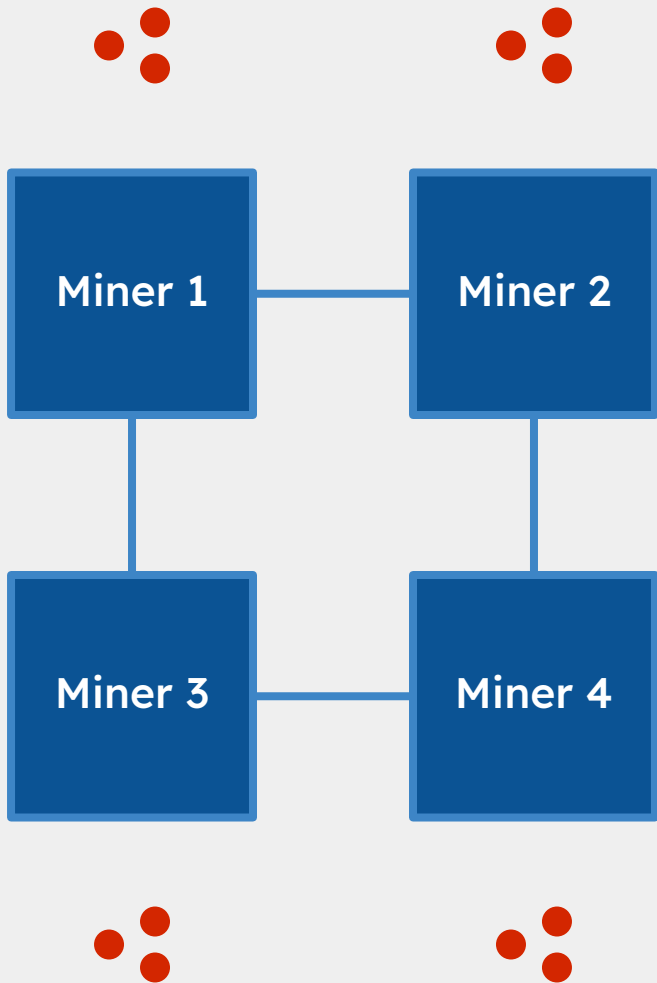
**Synced Blockchain**



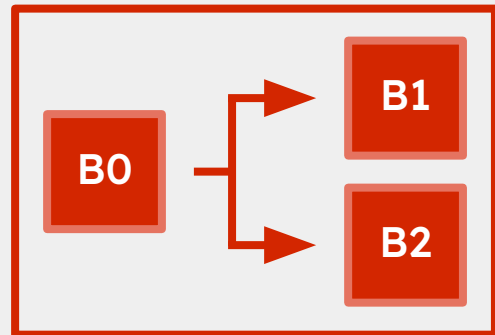


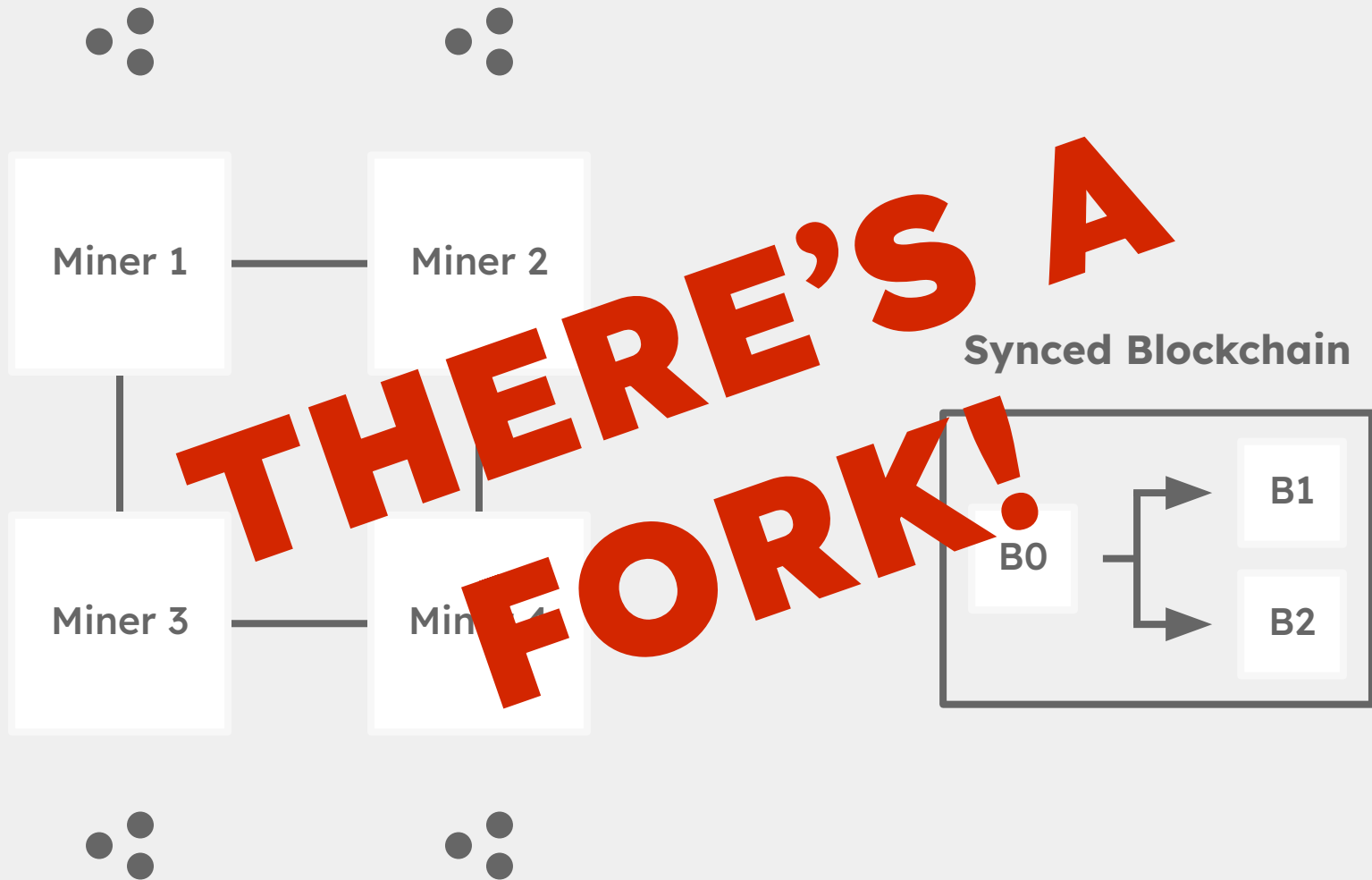
**Synced Blockchain**

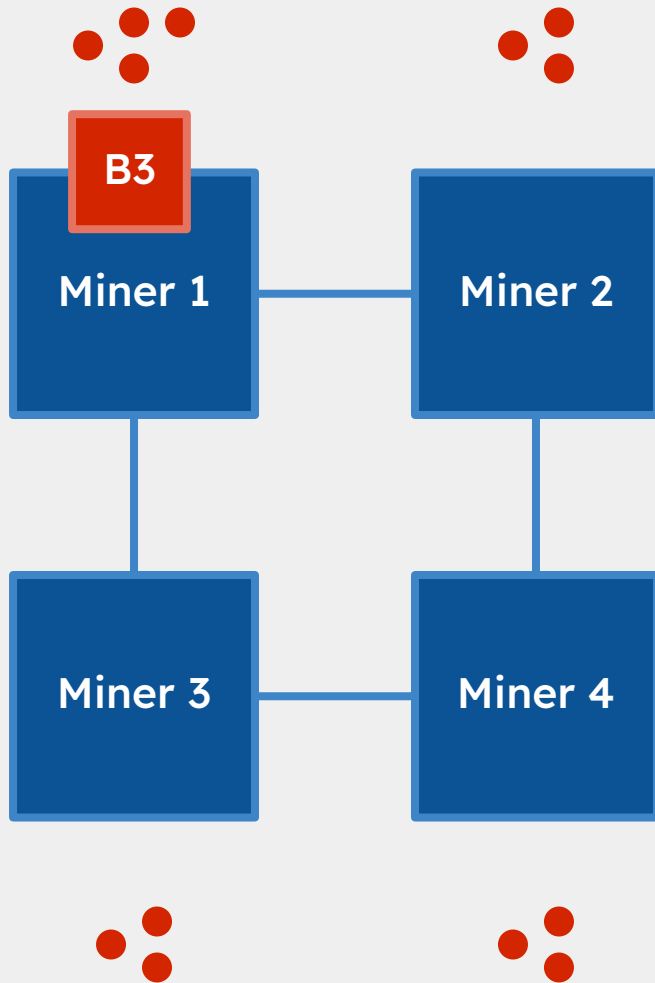




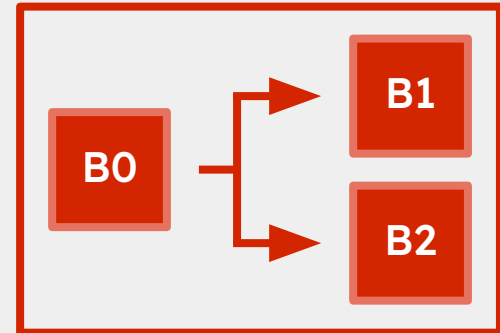
## Synced Blockchain

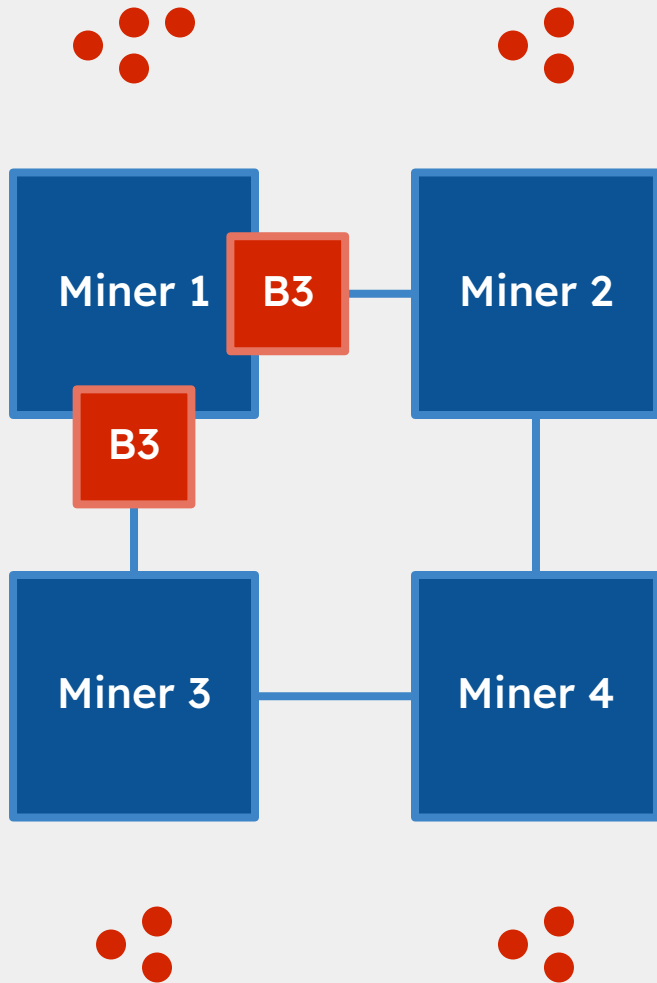




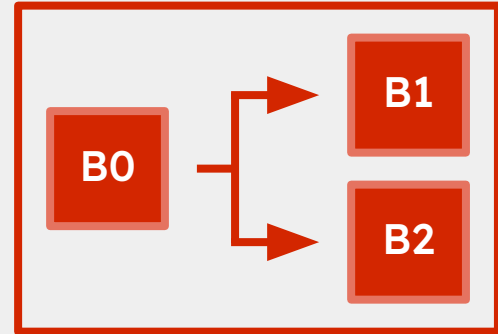


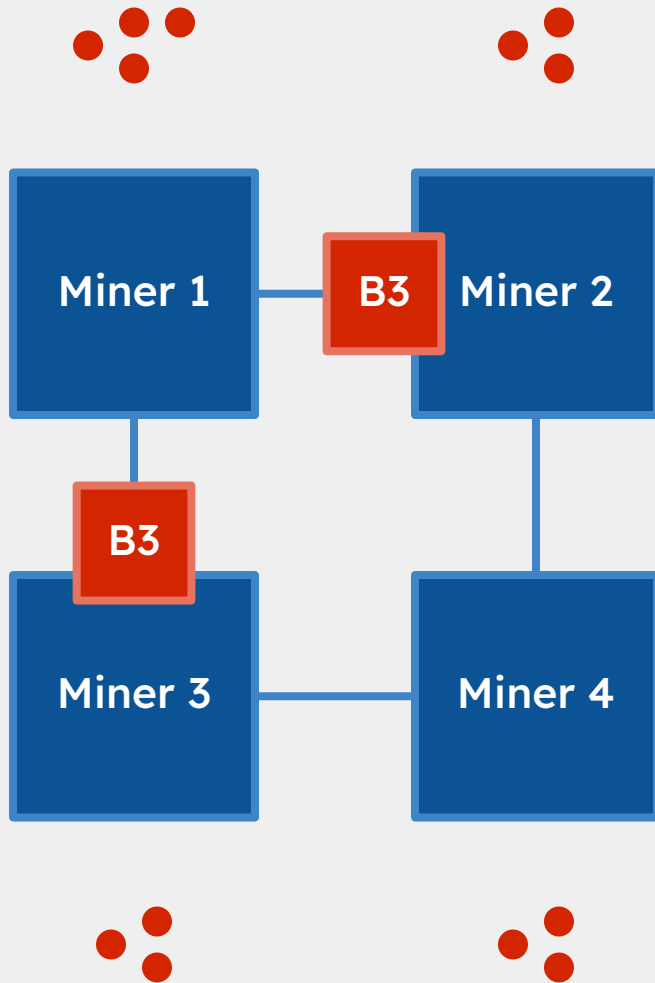
## Synced Blockchain



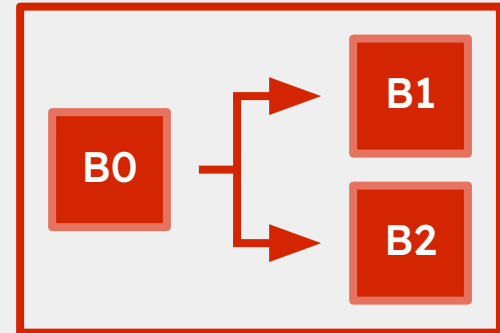


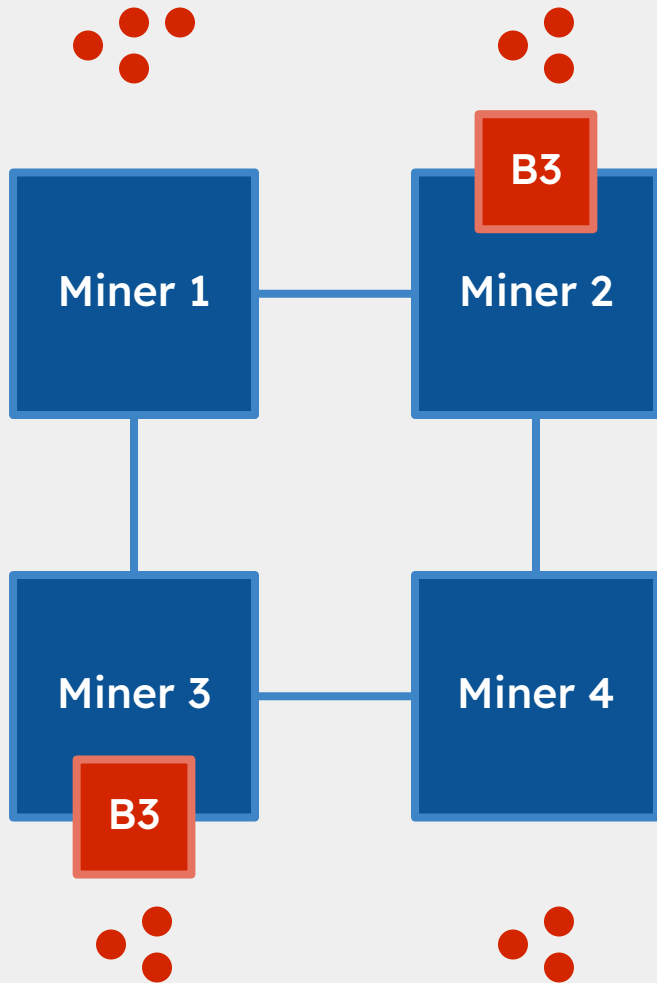
## Synced Blockchain



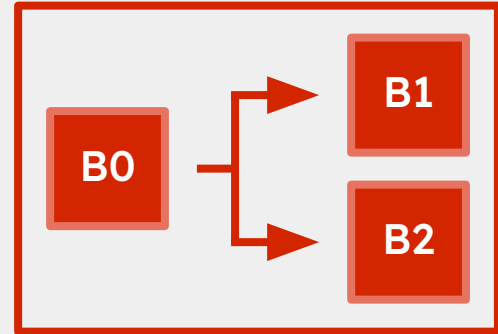


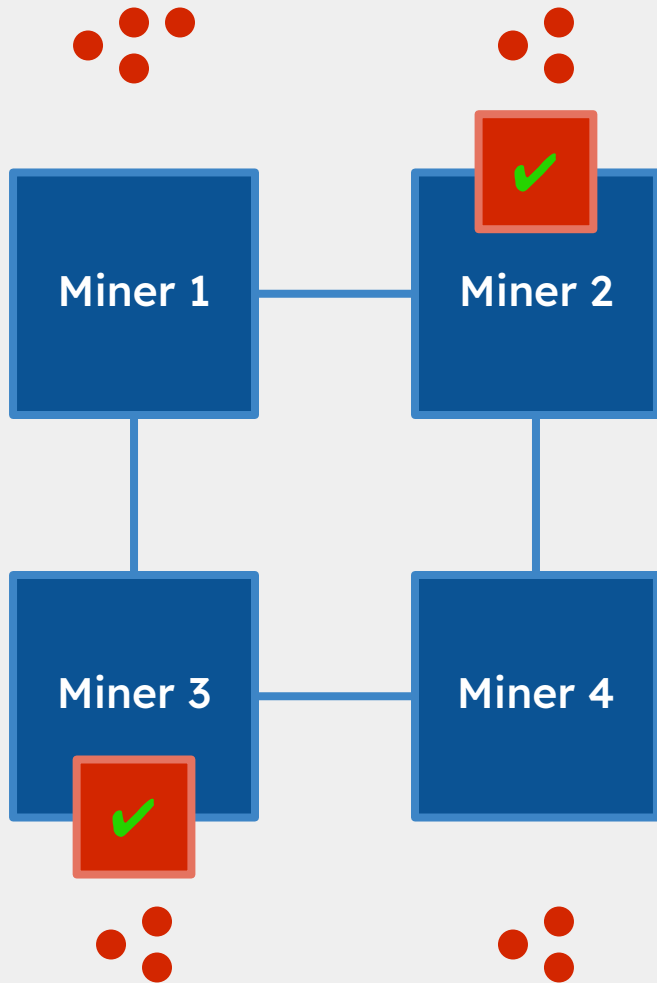
## Synced Blockchain



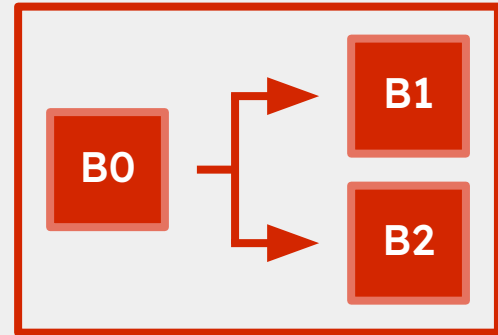


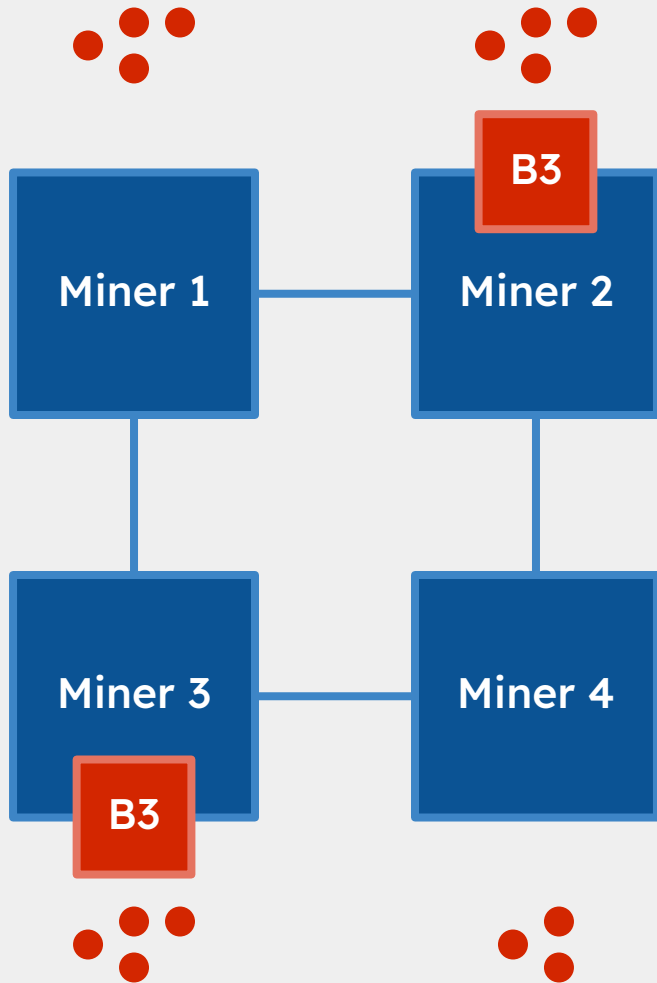
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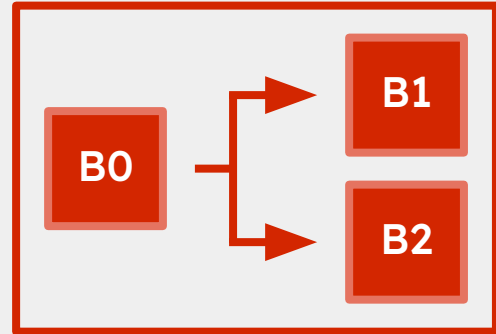


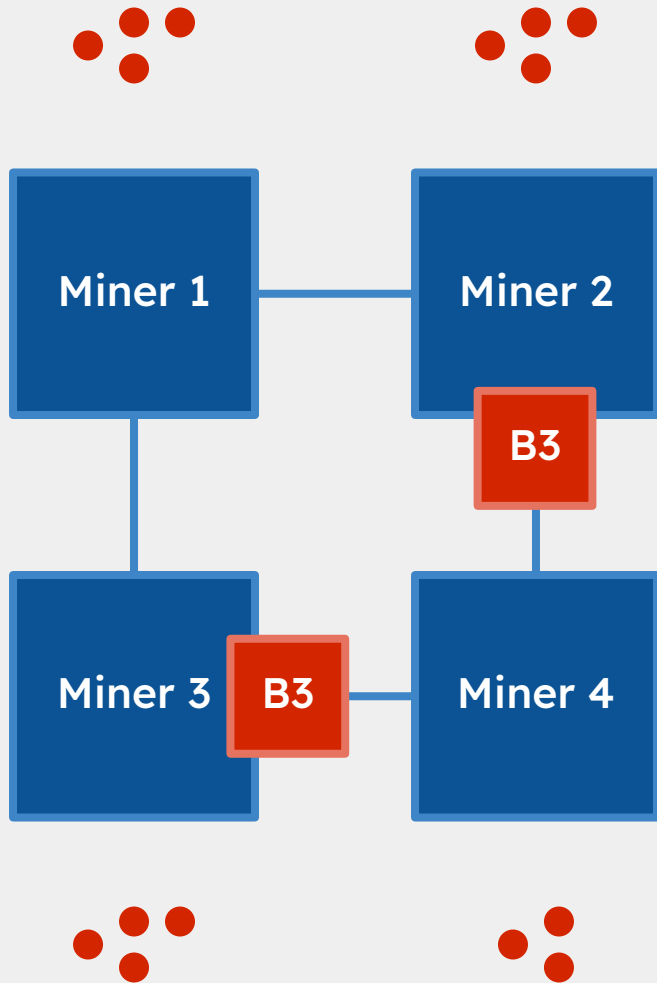
## Synced Blockchain



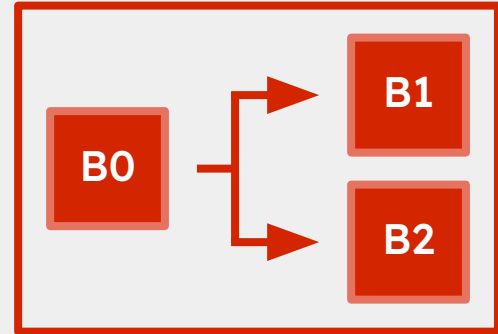


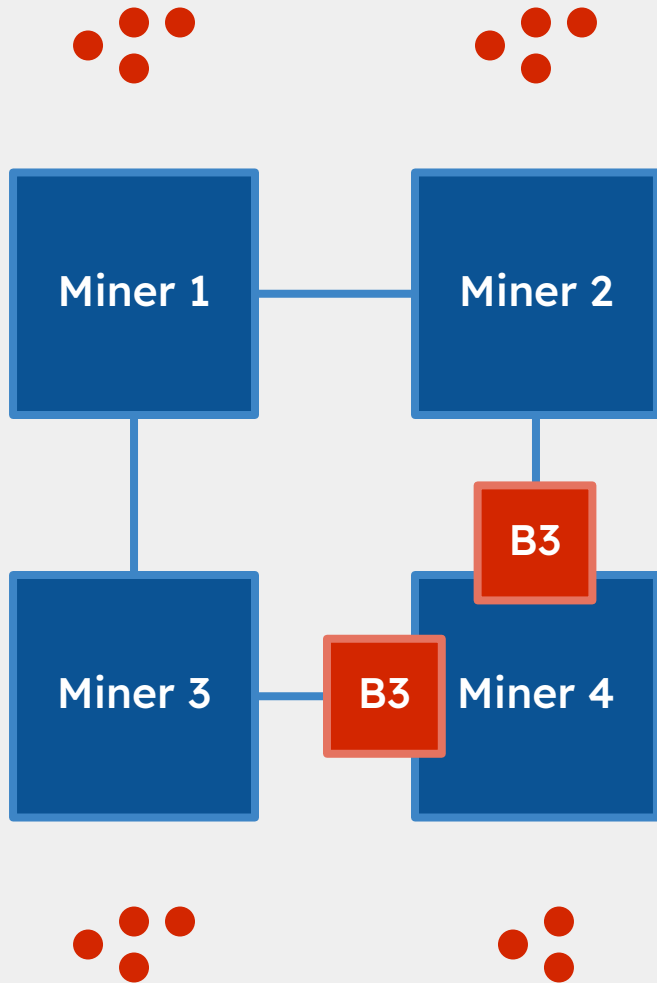
## Synced Blockchain



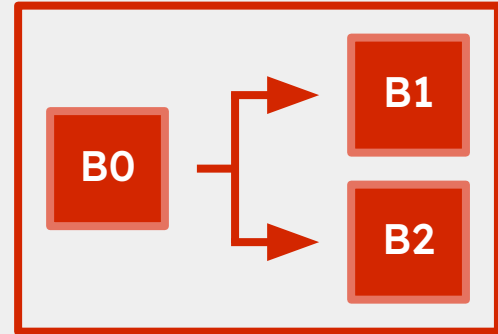


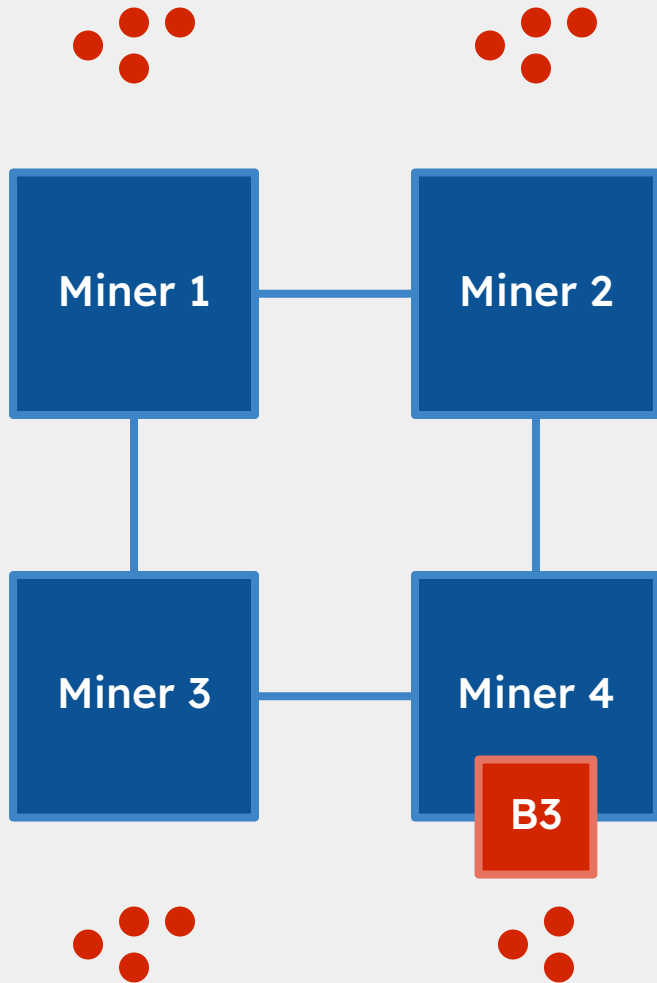
## Synced Blockchain



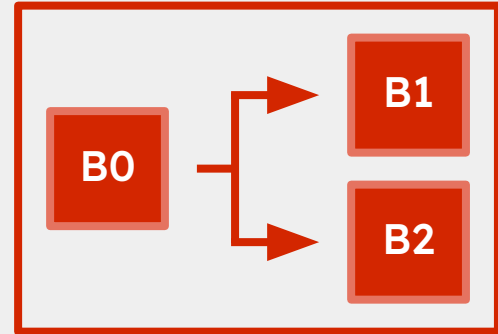


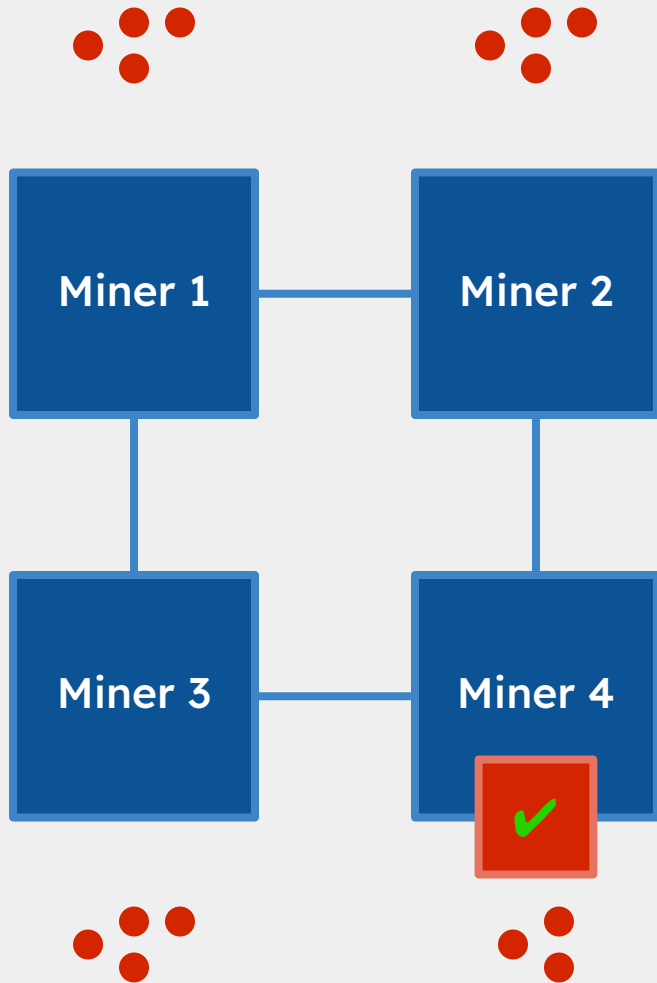
## Synced Blockchain



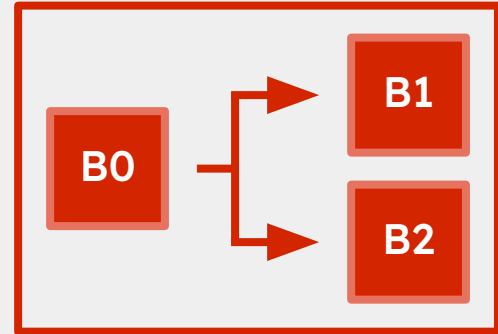


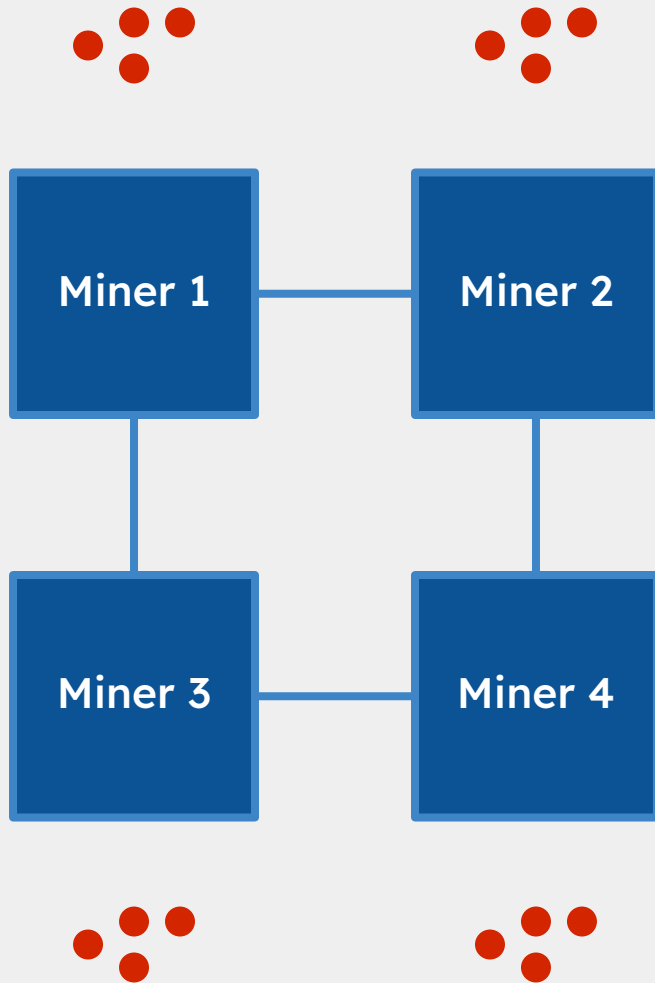
## Synced Blockchain



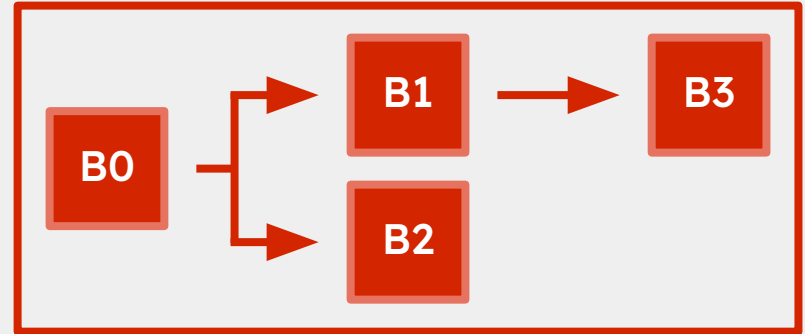


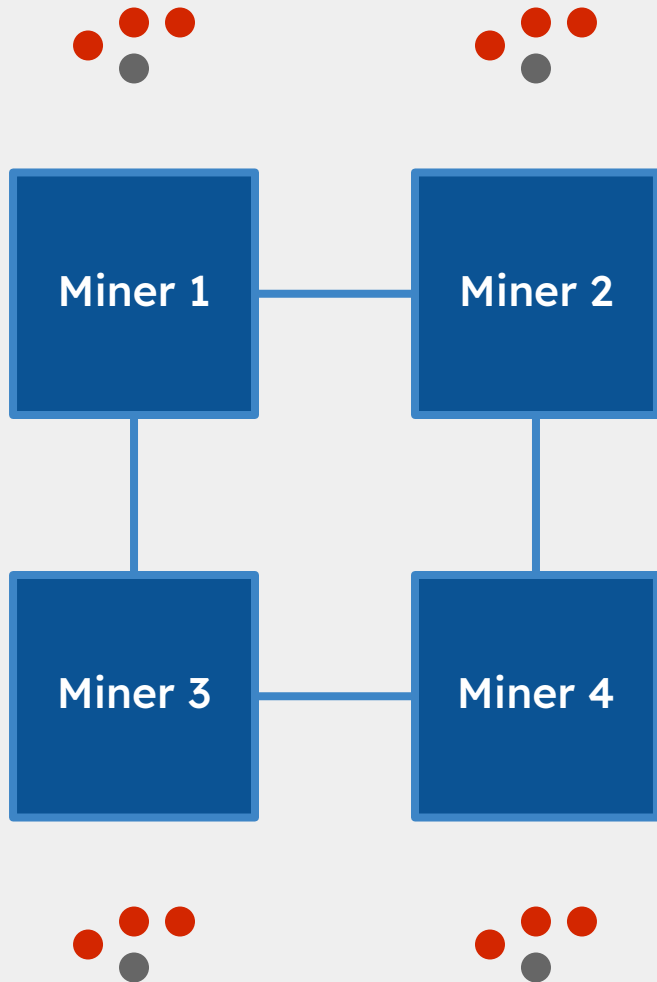
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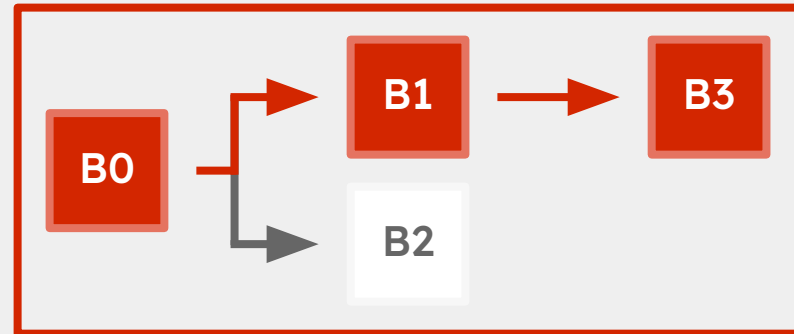


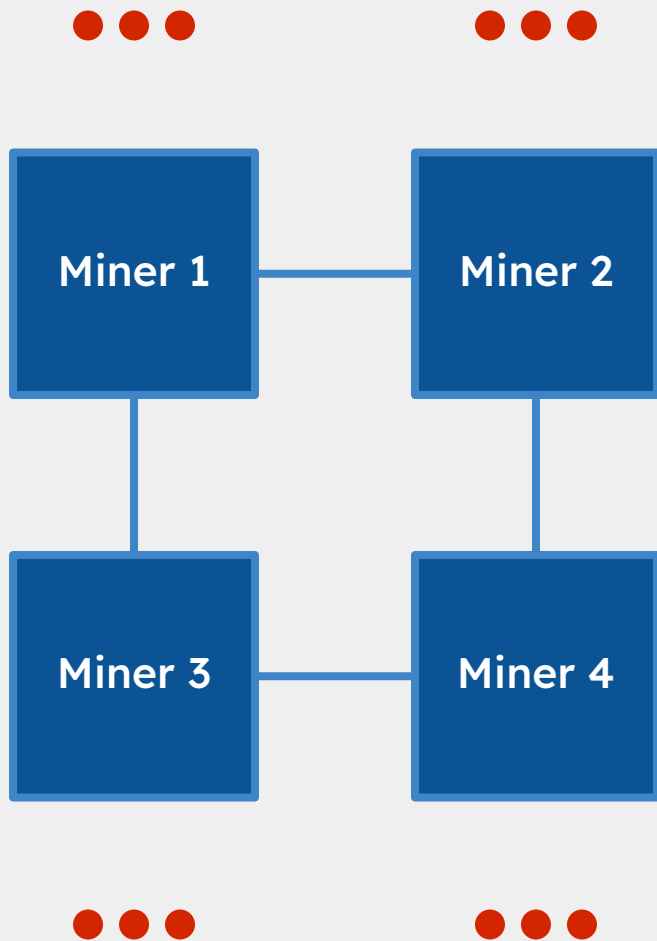
## Synced Blockchain





## Synced Blockchain





### Synced Blockchain



...

...

Miner 1

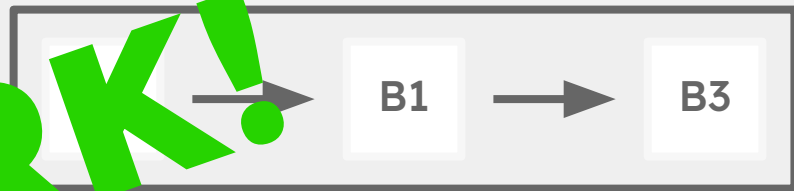
Miner 2

Miner 3

Miner 4

**NO MORE  
FORK!**

Synced Blockchain



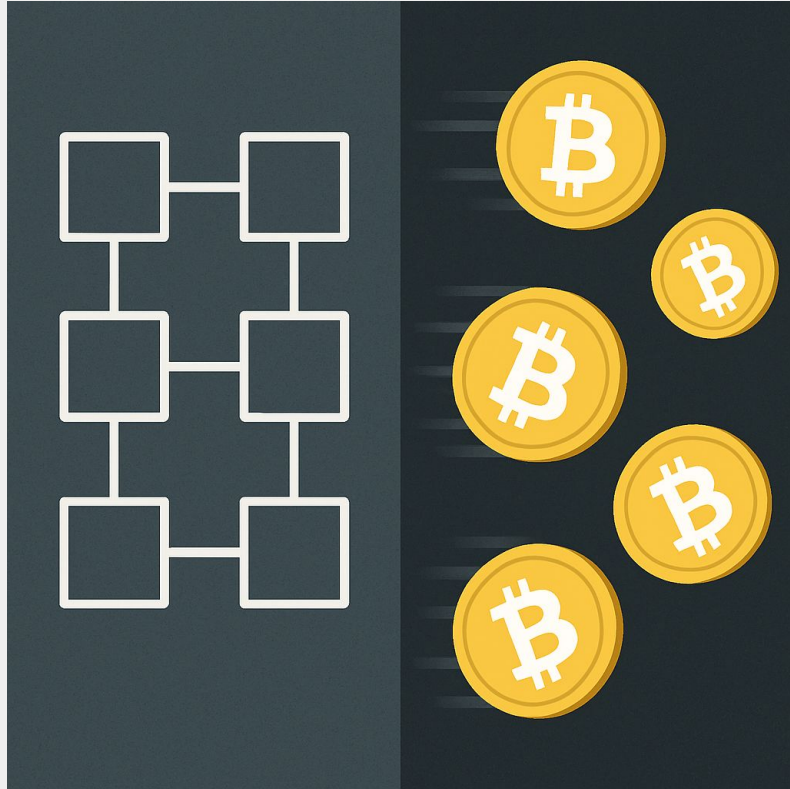
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...



# Transactions, Transactions

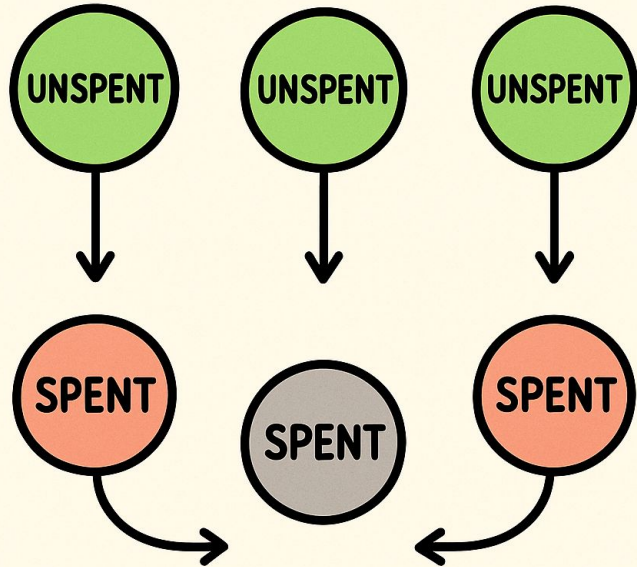
# Why Transactions Matter?



- Blocks record history; transactions move value
- Today: Bitcoin's open model vs Monero's privacy
- Same math, opposite privacy goals

# UTXO Model Basics

## COIN OUTPUTS



- Each coin sits in exactly one UTXO
- Spending “smashes” old outputs, creates new ones “
- Your balance = sum of your UTXOs

# Anatomy of a Bitcoin Tx

## TX PACKET

|         |        |         |          |
|---------|--------|---------|----------|
| VERSION | INPUTS | OUTPUTS | LOCKTIME |
|---------|--------|---------|----------|

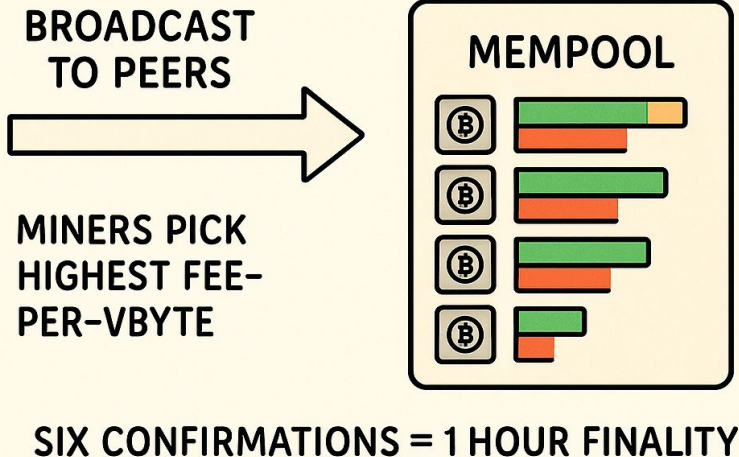
$$\sum \text{inputs} = \sum \text{outputs} + \text{fee}$$

$$\text{TxID} = \text{SHA-256}^2_{(\text{transaction})}(\text{serialized})$$

- Four key fields: version | inputs | outputs | locktime
- Conservation:  $\sum \text{inputs} = \sum \text{outputs} + \text{fee}$
- $\text{TxID} = \text{SHA-256}^2(\text{serialized transaction})$

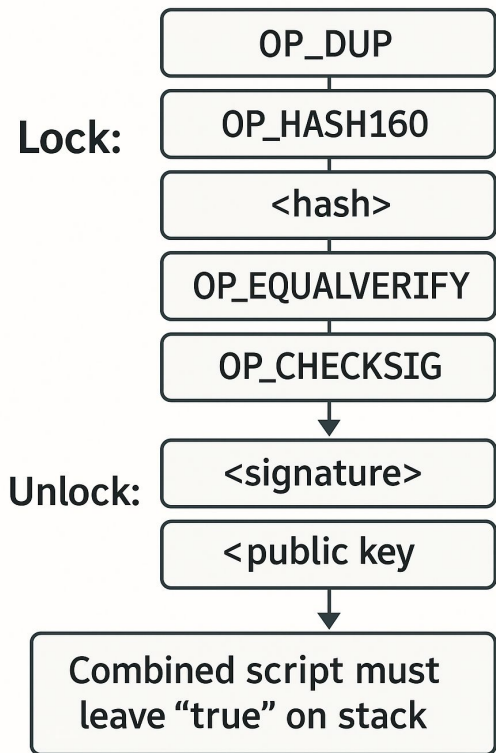
# Getting Into the Mempool

## GETTING INTO THE MEMPOOL



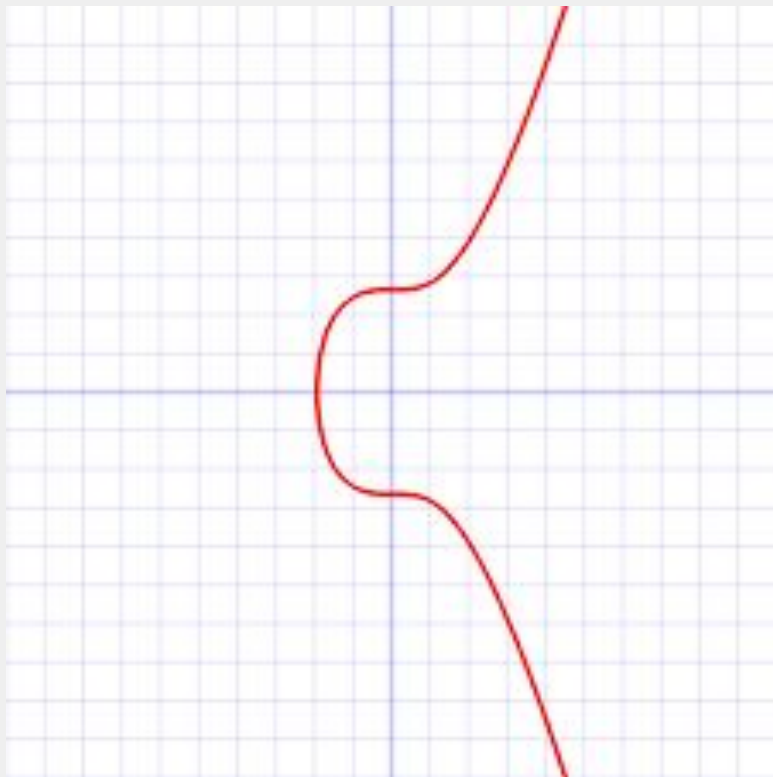
- Broadcast to peers → mempool
- Miners pick highest fee-per-vbyte
- Six confirmations  $\approx$  1 hour finality

# Locking & Unlocking Scripts



- **Lock:** OP\_DUP OP\_HASH160  
<hash> OP\_EQUALVERIFY  
OP\_CHECKSIG
- **Unlock:** <signature> <public  
key>
- Combined script must leave  
"true" on stack

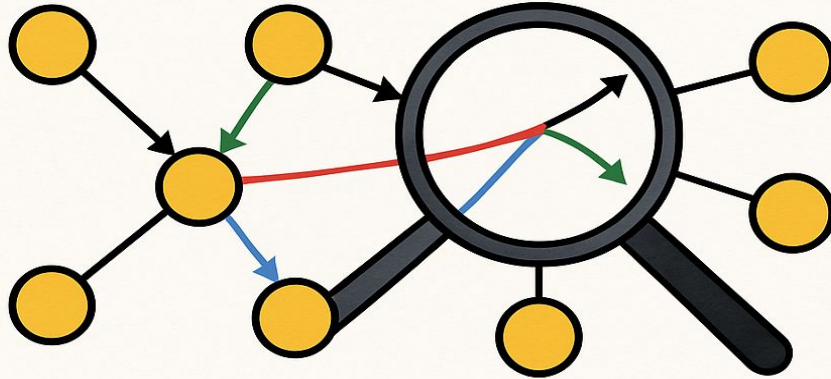
# ECDSA on secp256k1



- Sign: pick random  $k$ , compute  $R=kG \Rightarrow r$ , then  $s= k^{-1}(z+dr)$
- Verify: check  $s^{-1} \cdot (zG + rQ) = R$
- Security: discrete-log hardness

# Privacy Limits in Bitcoin

## PRIVACY LIMITS IN BITCOIN

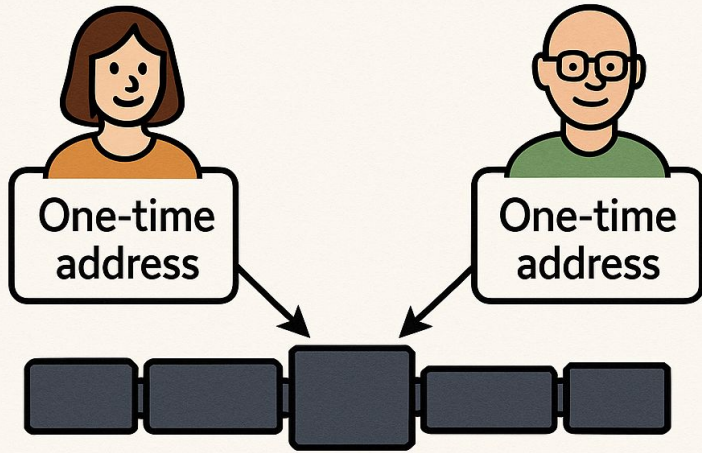


All UTXOs, addresses, and amounts  
are on-chain forever

Graph heuristics cluster related wallets

Real-world de-anonymization via  
KYC data

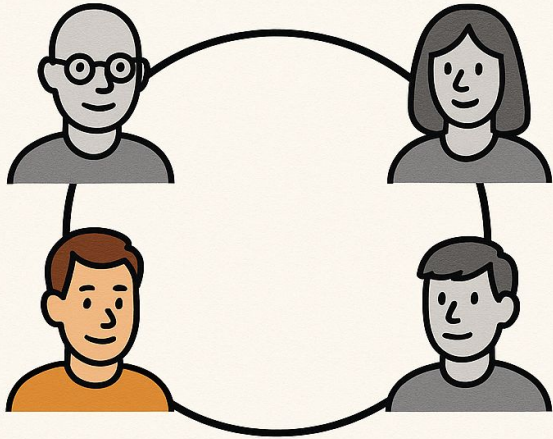
# Monero's Stealth Addresses



- Sender computes  $P = Hs(rAp)G + Bp$
- Blockchain only shows  $P$ , not  $Ap$  or  $Bp$
- View-key lets recipient scan and recover  $P$

# Ring Signatures (MLSAG)

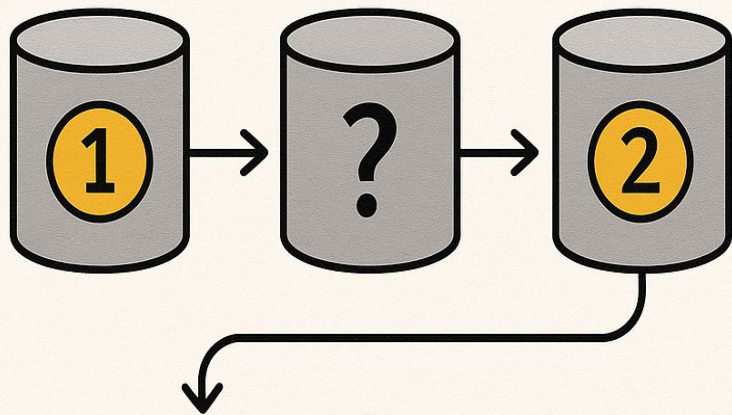
## Ring Signatures (MLSAG)



A ring of possible senders  
with one real signer highlighted

- Form a ring of  $N$  outputs (1 real +  $N-1$  decoys)
- MLSAG proof shows you know one secret key
- Key-image  $I = x \cdot H_p(P)$  prevents double-spending
-

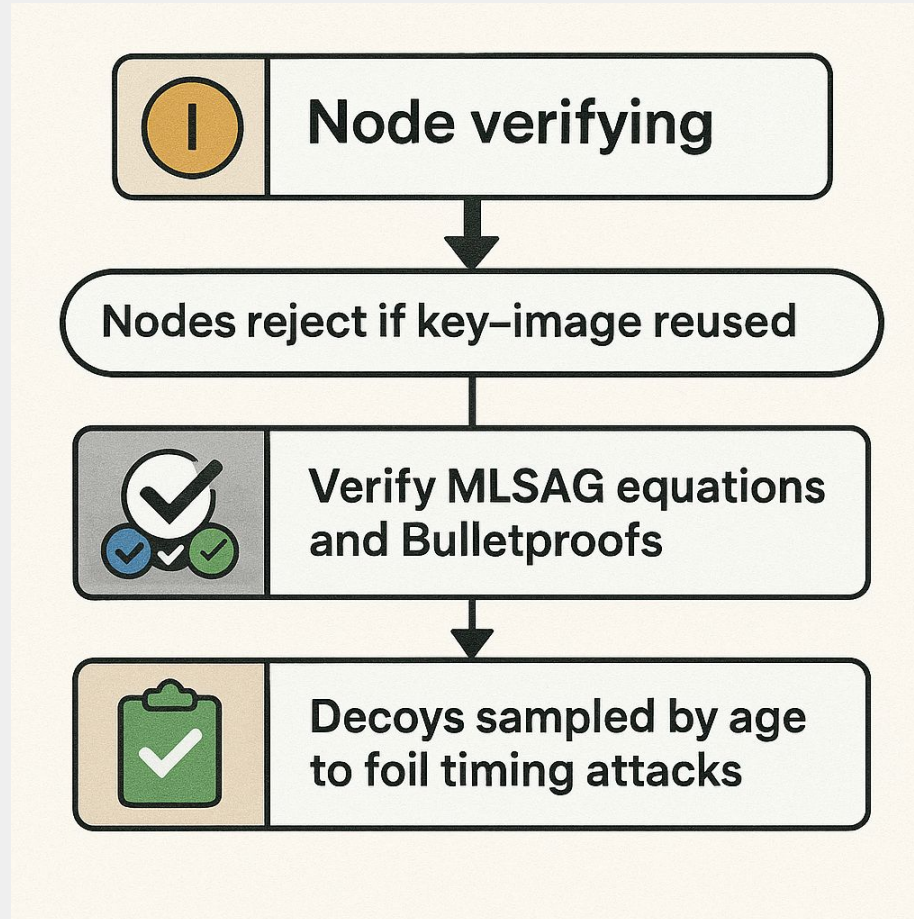
# RingCT & Hidden Amounts



Amounts concealed inside  
opaque commitments

- Amounts hidden in Pedersen commitments  $C = vG + bH$
- Bulletproofs supply zero-knowledge range proofs
- Net balance:  $\Sigma C_{in} - \Sigma C_{out} = f \cdot H$

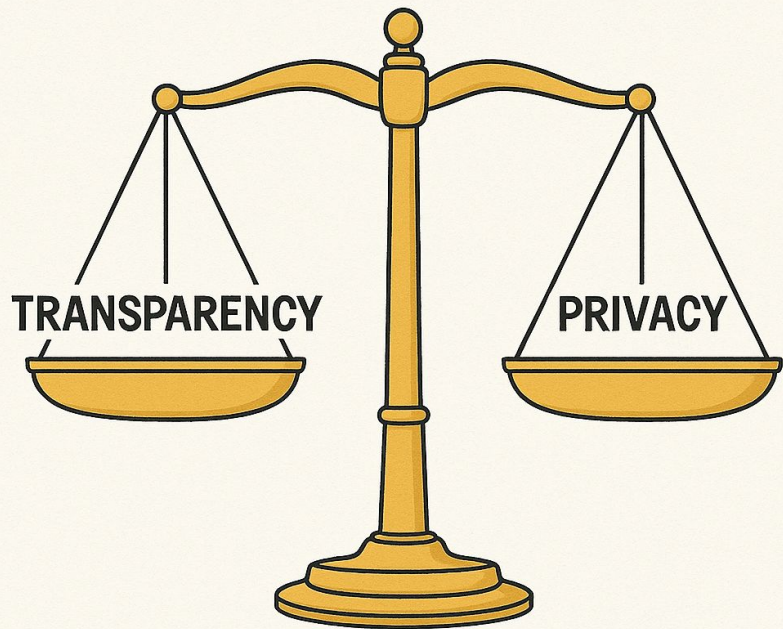
# Verification & Decoy Sampling



## Bitcoin vs Monero

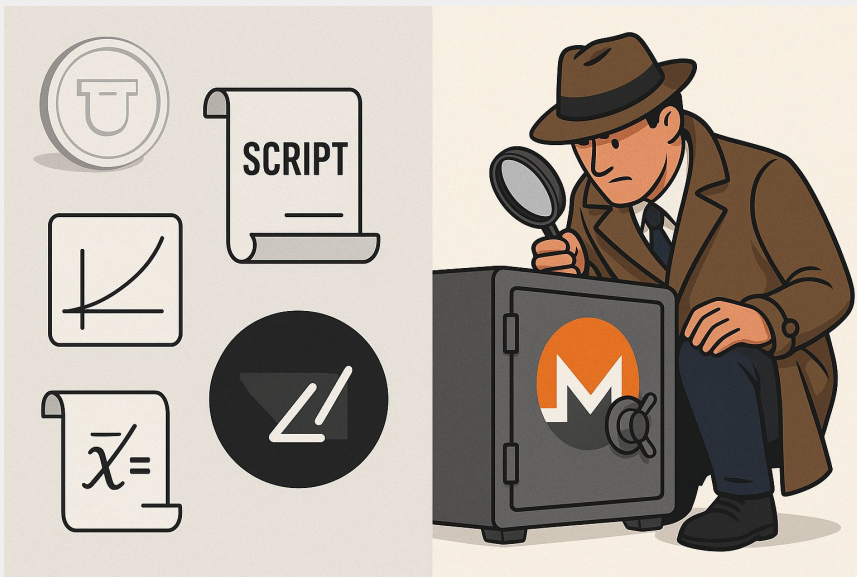
| Feature        | Bitcoin       | Monero        |
|----------------|---------------|---------------|
| Address reuse  | possible      | impossible    |
| Sender visible | yes           | no            |
| Amount visible | yes           | no            |
| Chain-analysis | deterministic | probabilistic |

# Security Trade-offs



- Transparency → easy auditing, faster sync, regulatory compliance
- Privacy → stronger anonymity, larger data, compliance friction
- Choose based on use-case priorities

# Key Takeaways & What's Next



- Four key fields: version | inputs | outputs | locktime
- Conservation:  $\sum \text{inputs} = \sum \text{outputs} + \text{fee}$
- $\text{TxID} = \text{SHA-256}^2(\text{serialized transaction})$



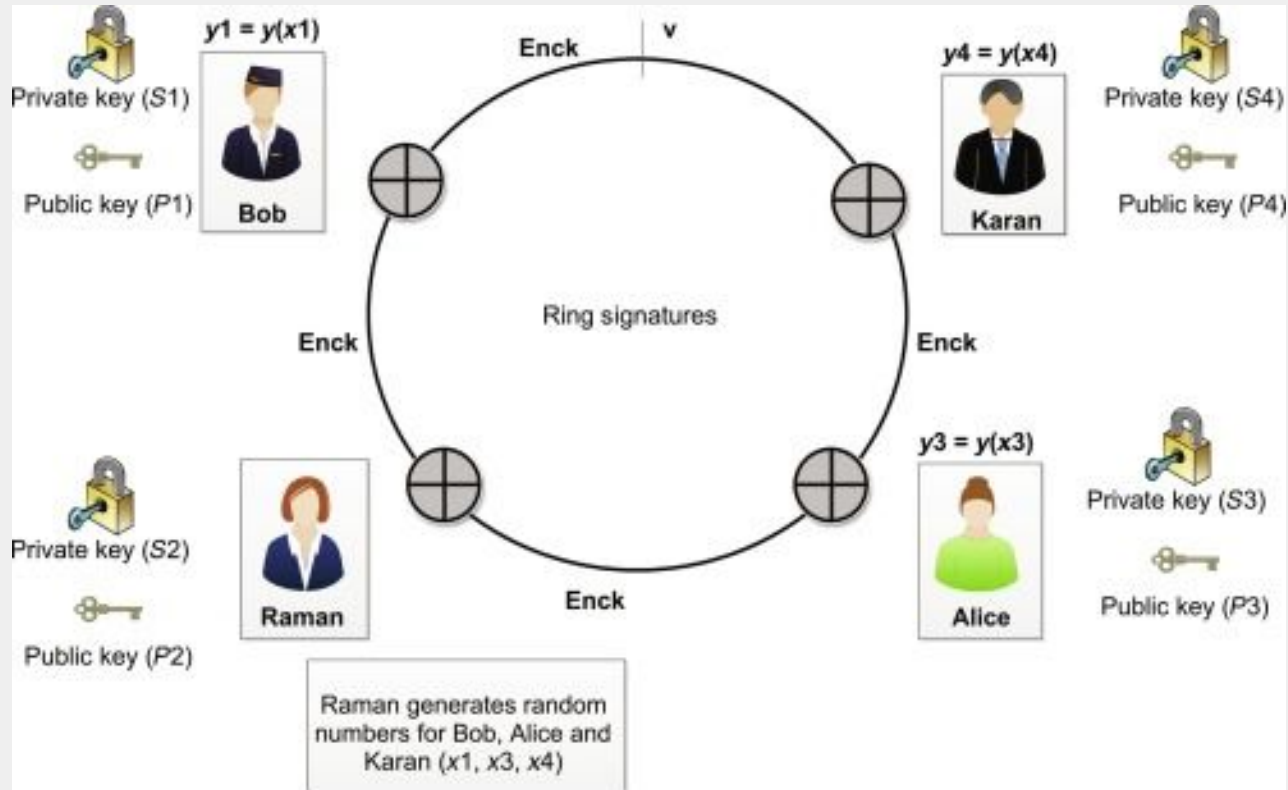
# Monero, and How To Break It

# How Monero Protects Privacy

3.1



# Ring Signatures



# RingCT (Confidential Transactions)



**Alice**

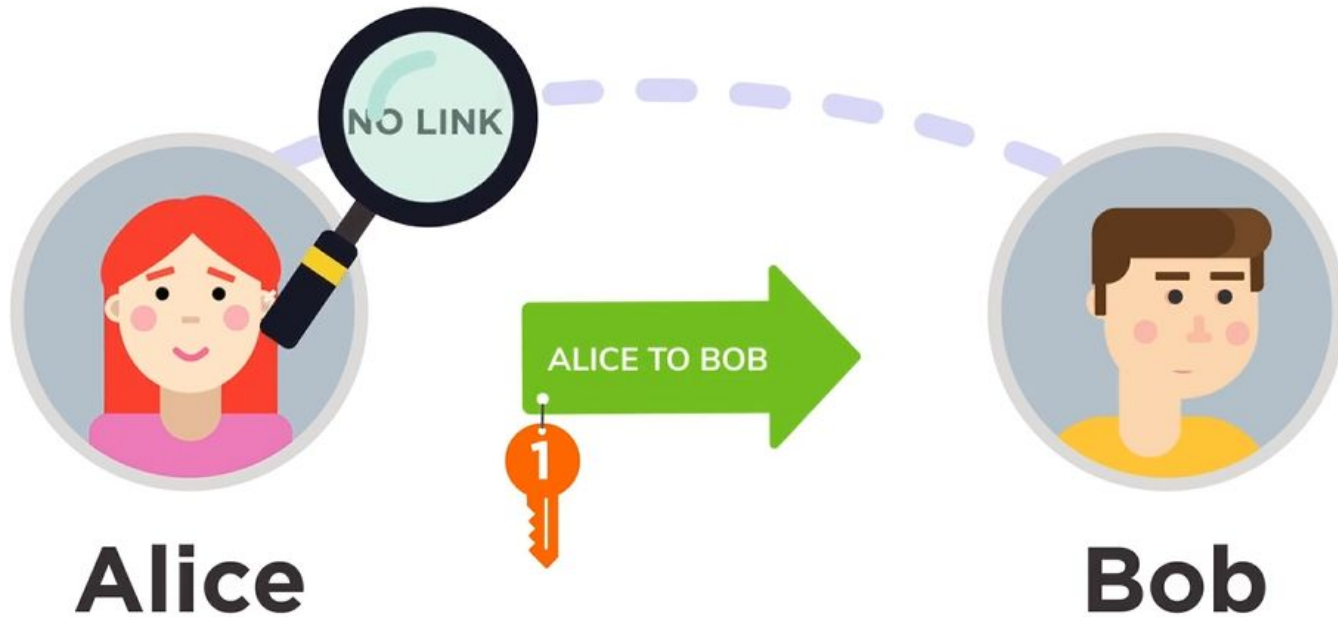
**commitment**



$$(rct = xG + aH(G))$$



# Stealth Addresses



# Dandelion++



# Based Monero



## MONERO EXPLAINED

Monero is digital cash. It's like Bitcoin, but your activity is kept confidential.



### FRESH CODEBASE

Written from the ground up to deliver genuine privacy and anonymity. Not just another Bitcoin clone.



### ALWAYS-ON PRIVACY

All transactions are confidential, which means every Monero user's activity enhances the privacy of other users.



### CONFIDENTIAL AMOUNTS

Transaction amounts are encrypted to prevent transactions from being traced by the amount sent.



### DYNAMIC BLOCKSIZE

Transaction volume is not restricted by an arbitrary cap, ensuring transactions are processed quickly and inexpensively.



### VIEW KEY AUDITING

Dual key system allows you to provide other computers or people with the ability to view your incoming funds without risk of theft.



### LIGHTNING MEMORY MAPPED DATABASE

Built on an ultra high performance, low memory footprint database developed by a core Monero contributor who worked on the NASA space shuttle.



### LIGHT WALLETS HARDWARE WALLETS

Monero supports both light wallet technology and will be available on future hardware wallet technology.



### MONERO RESEARCH LAB

World class academic researchers that are pushing the frontier of what is possible with cryptocurrency privacy and security.



### FUNGIBILITY

Monero's untraceability means that you do not have to fear that coins you receive are tainted by association with the activity of their previous owners.

### STEALTH ADDRESSES

Your wallet address never appears on the public blockchain. This means no one that knows your wallet address can see how rich you are or how you spend.



### UNTRACEABLE

The source of funds for each transaction is concealed using established and proven ring signature cryptographic techniques. This is achieved without the need for a trusted setup, safeguarding against malicious currency fabrication.



### HIGH SECURITY ED25519 CURVE ENCRYPTION

Cutting edge Elliptic Curve Cryptography that protects your funds from theft.



### ASIC RESISTANCE

CryptoNight mining algorithm is resistant to special-purpose hardware that would result in unhealthy mining centralization.



### NETWORK INVISIBILITY

Monero's Kovri project encrypts and routes transactions through the I2P Invisible Internet Project nodes to protect you from network monitoring.



### RAPID, EFFICIENT NODE INTERCONNECT

High performance, high scalability messaging protocol for node communication.



### OPENALIAS IDENTITY

A Monero project that makes paying someone else as easy as knowing their email address (no complicated strings of random numbers and letters required).



### GRASSROOTS COMMUNITY

Over 240 developers have contributed to the Monero project. Forums and chat channels are welcoming and active.



### EXCHANGE TRADED

Accepted widely at exchanges, allowing you to get your funds in and out of Monero easily.



# Real-World Use of Monero

3.2



# Ransomware (e.g. REvil and DarkSide)

**Your computer has been infected!**



Your documents, photos,  
databases and other important files  
encrypted



To decrypt your files you need to  
buy our special software -  
[REDACTED]-Decryptor



Follow the instructions below. But  
remember that you do not have  
much time

[REDACTED]-Decryptor price

You have **6 days, 23:59:40**

\* If you do not pay on time, the price will be doubled

\* Time ends on Jul 9, 19:09:53

Current price **213.23496134 XMR**  
≈ 44,999 USD

After time ends **426.46992268 XMR**  
≈ 89,998 USD

Monero address: [REDACTED]

\* XMR will be recalculated in 5 hours with an actual rate.

# Darknet Markets (e.g. Alpha Bay)

Home | AlphaBay Market - Tor Browser

Home | AlphaBay Market

← → ×

alphabay522szl32u4ci5e3iokdsyth56ei7wngr2wm7i5jo54j2eid.onion/homepage

⌵ ⋮ ☆ 🛡️ 🔍 ☰

 **AlphaBay Market**

HOME FORUM ROADMAP BLOG MY CABINET ▾ NOTIFICATIONS (0)

Logged in as ██████████  
Balance: 0.00000000 XMR  
[My Cabinet](#) | [Support](#) | [Logout](#)

Home

XMR / USD 233 BTC / USD 41583.26

Public Profile



Name: ██████████

Joined: 09/29/2021

Last Login: 09/29/2021

Total Orders: USD 0.00

Trust Level: Level 1

Categories

██████████

██████████

██████████

 **Welcome,** ██████████

AlphaBay is **re-opened for business** for the first time exactly 4 years since 2017. Having first **started trading in 2014**, we cultivated a strong connection with the DNM community by being **professional, fair & responsive** when dealing with buyers and sellers alike. By 2017 that alongside our innovations, high security, uptime & ease of use made us **the biggest and most trusted secure anonymous marketplace** ever to have existed. You can read our [first public message since 2017 detailing our vision going forward, why we have returned, what has been improved and more answers here.](#)

If the name AlphaBay does not ring a bell and you do not know what this website does or who we are, you should search us up yourself. You can also [view our page about AlphaBay here.](#)

AlphaBay has always been a **principle-driven, vision-oriented and community-centered** marketplace. The administration is almost the same as in the 2014-2017 period which means you will find again AlphaBay runs with **mature management, unparalleled security, 24/7 professional & well-trained Staff** and of course unique vision for the future not only for itself but [for the darknet market scene](#) too.

If you have any questions about the market refer to [FAQ](#), [How-To guides](#) or [Support Center](#). Feedback or suggestions can be directed to [the Forum](#) and our Staff members will look into it. We greatly appreciate any feedback you can provide to make the market a better place!

 **Quick Search**

Transferring data from alphabay522szl32u4ci5e3iokdsyth56ei7wngr2wm7i5jo54j2eid.onion...

# Cryptojacking



# Can Monero Be Traced?

3.3



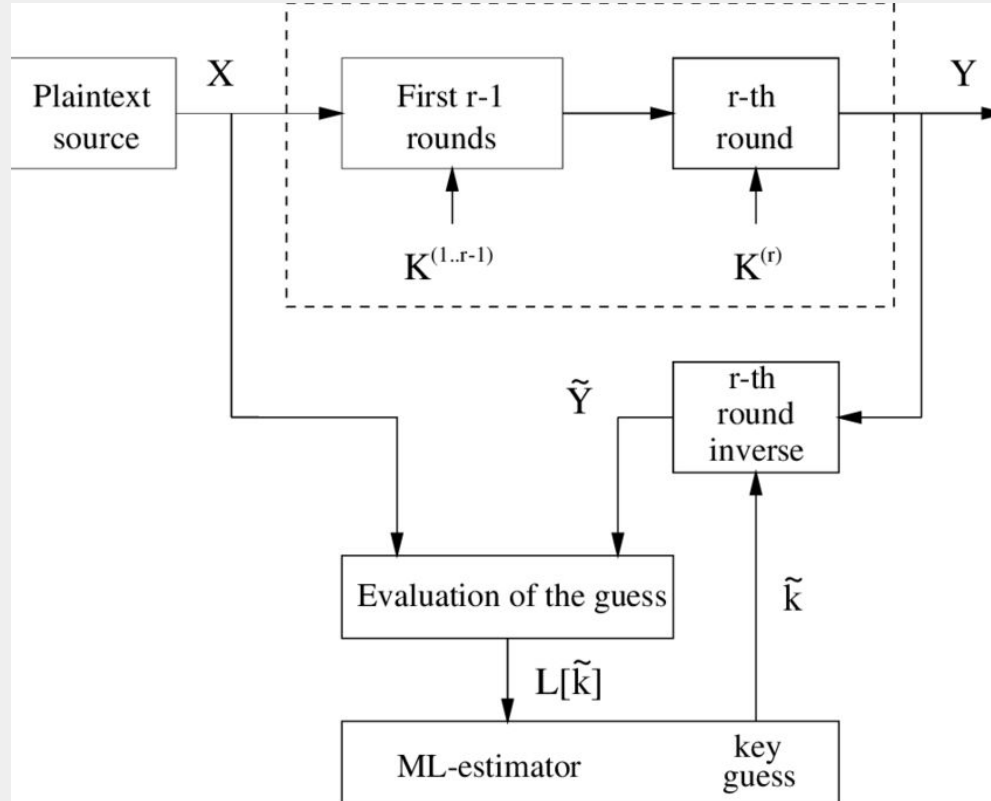
# IRS Bounty (\$625,000)



# Chainalysis & CipherTrace



# Statistical attacks (Ring reuse, timing)



# User & exchange leaks

**James Edwards @librehash** Mar 20

1/ Not sure how this space missed this but **@fluffypony** (Riccardo Spagni), the founder & lead maintainer of **@monero \$XMR** outed himself as an informant for Interpol after he was detained by U.S. Marshals pending extradition. Evidence strongly suggests he helped them track Monero.

**Monero Maintainer 'Fluffypony' Arrested and Extradited for Non-Crypto Crimes**

Monero's lead maintainer, Riccardo Spagni, who was arrested in Tennessee, will be extradited to South Africa to face charges.

Aug 2, 2021 at 6:07 pm EDT Updated Sep 14, 2021 at 9:34 am EDT

Known in the digital currency world as Fluffypony for five years, Spagni was arrested in the U.S. after helping a tech company track down a South African hacker.

However, the maintainers of the Monero blockchain have stated that he is not a hacker.

in court that he is

coingeek

business 10 June 2020

**Monero's Riccardo Spagni accused of fraud and forgery in South Africa**

The South African government has accused digital currency project Monero's lead maintainer, Riccardo Spagni, of fraud and forgery in South Africa.

Spagni is accused of defrauding a South African company of \$15,000 a decade ago.

Known in the digital currency world as Fluffypony for five years, Spagni was arrested in the U.S. after helping a tech company track down a South African hacker.

As South African prosecutors have charged Fluffypony with fraud and forgery.

However, the maintainers of the Monero blockchain have stated that he is not a hacker.

in court that he is

2023 8:03 PM UTC

55,164 ❤️ 1,156



# Privacy vs. Regulation

3.4



# Exchange Delistings



# Insurance won't cover Monero ransoms



# Government Crackdowns

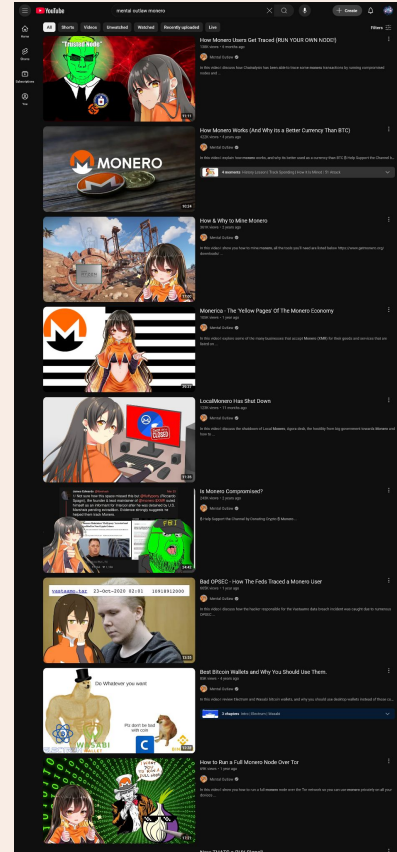


# Uphill Battle



# Mental Outlaw on YouTube

3.5





# Crypto Concerns and Scams

# Pseudonymity is not Anonymity

4.1



Anonymity



Pseudonymity



Real-Life Identity

**BLOCKCHAIN**[WALLET](#)[CHARTS](#)[STATS](#)[MARKETS](#)[API](#)

 Search for block hash, transaction, address, etc

### LATEST BLOCKS

[SEE MORE →](#)

| Height                 | Age        | Transactions | Total Sent    | Relayed By              | Size (kB) | Weight (kWU) |
|------------------------|------------|--------------|---------------|-------------------------|-----------|--------------|
| <a href="#">488256</a> | 7 minutes  | 646          | 2,507.44 BTC  | <a href="#">BTC.com</a> | 307.49    | 1,187.07     |
| <a href="#">488255</a> | 12 minutes | 979          | 7,085.24 BTC  | <a href="#">BTC.com</a> | 455.69    | 1,714.29     |
| <a href="#">488254</a> | 17 minutes | 1255         | 7,667.10 BTC  | <a href="#">BitFury</a> | 960.12    | 2,774.96     |
| <a href="#">488253</a> | 25 minutes | 2070         | 18,839.79 BTC | <a href="#">AntPool</a> | 994.12    | 3,868.78     |

### NEW TO DIGITAL CURRENCIES?

Like paper money and gold before it, bitcoin and ether allow parties to exchange value. Unlike their predecessors, they are digital and decentralized. For the first time in history, people can exchange value without intermediaries which translates to greater control of funds and lower fees.

[BUY BITCOIN →](#)[LEARN MORE →](#)[GET A FREE WALLET →](#)

### SEARCH

You may enter a block height, address, block hash, transaction hash, hash160, or ipv4 address...

 Address / ip / SHA hash

Search

All transactions are available to view publicly. All it takes to have your transaction history fully known is to link your identity to your wallet address.

# Wallet Association Analysis



Transaction

Find links between wallets using transaction history

# Cross-Referencing Public Information

BLOCKCHAIN.COM

ProductsDataExplorer

Q

LoginSign Up

Bitcoin

\$8,806.89

Blocks

Transactions

Average Fee

Average Value

Difficulty

Hashrate

Latest unconfirmed transactions

View more transactions

| Hash                                              | Time  | Amount (BTC)   | Amount (USD) |
|---------------------------------------------------|-------|----------------|--------------|
| a9d16676c2b8f606d13eae75878f5bec7114429b6c88...   | 17:35 | 0.00272511 ... | \$24.00      |
| 1431f5dd7db527ede01ace4183764731e47eff3dbecbf1... | 17:35 | 0.02067509...  | \$182.08     |
| ee89d9e175bb6039a544ed3aca0885d8495af20a2b4...    | 17:35 | 0.09099077...  | \$801.35     |
| 33e77d5bbe51bec3dddf161604fae7ba25667572615b...   | 17:35 | 0.01247258...  | \$109.84     |
| 69b42dabc43c30b19119a3553a334ad3da72d3c4410...    | 17:35 | 0.11144077 ... | \$981.45     |

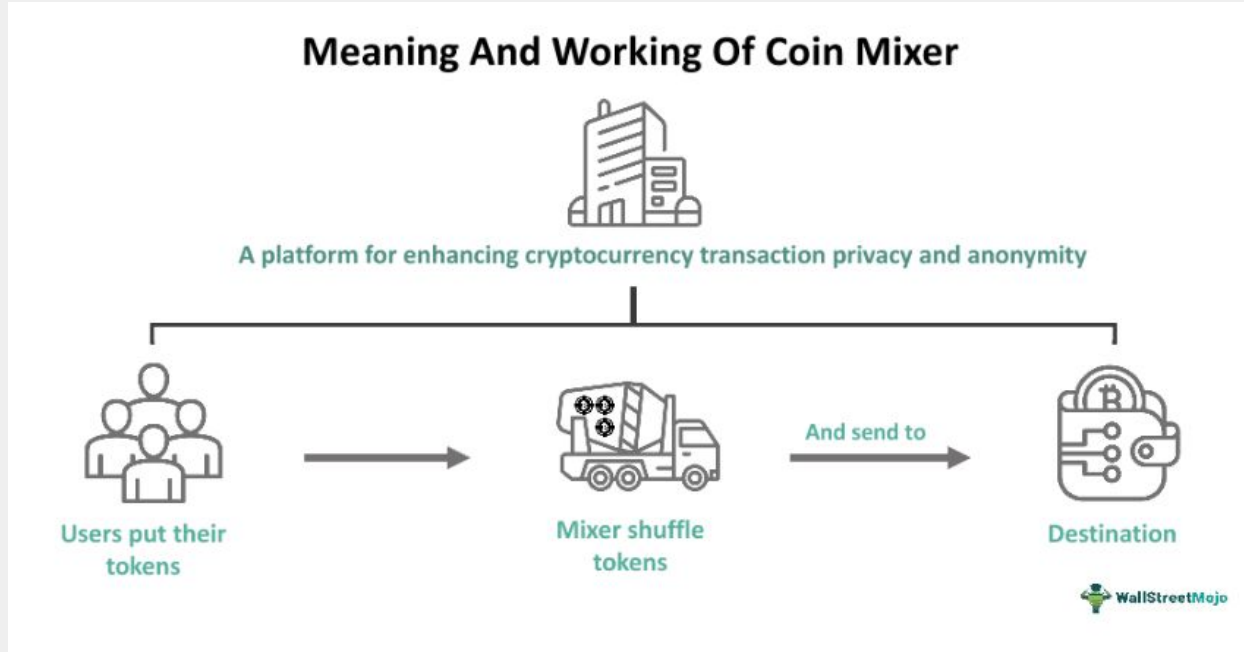
Ex. If someone does a public giveaway for .1144077... BTC, you can look at transaction history to find likely wallet candidates

# Know Your Customer (KYC)



If you use a large, registered exchange, you are obligated to give personal information to said exchange to verify your identity

# Solutions Exist, but...

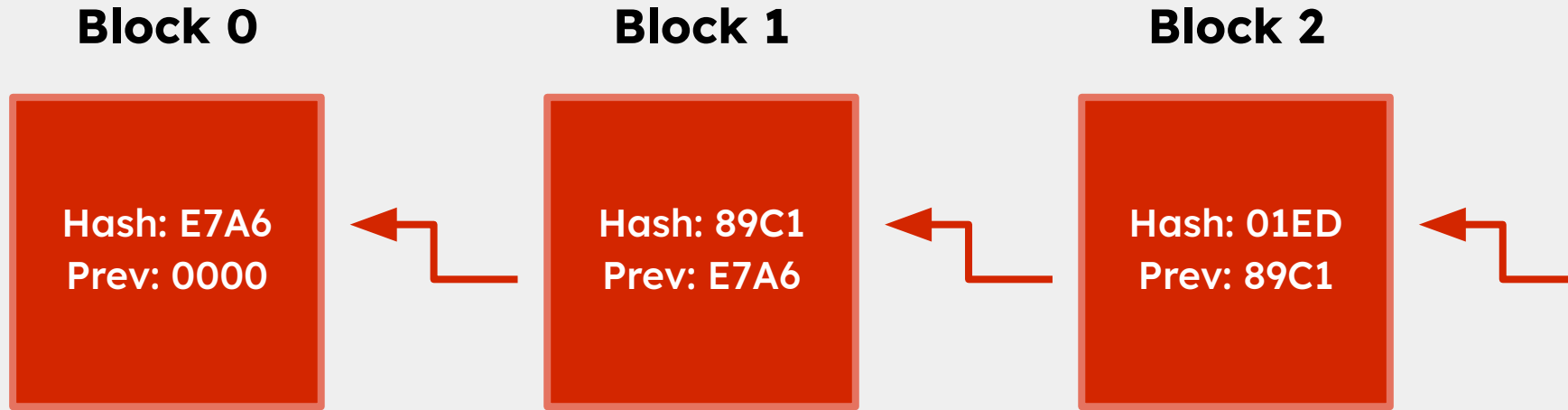


They can range from “difficult” to “probably illegal”, and one small mistake leaves you vulnerable

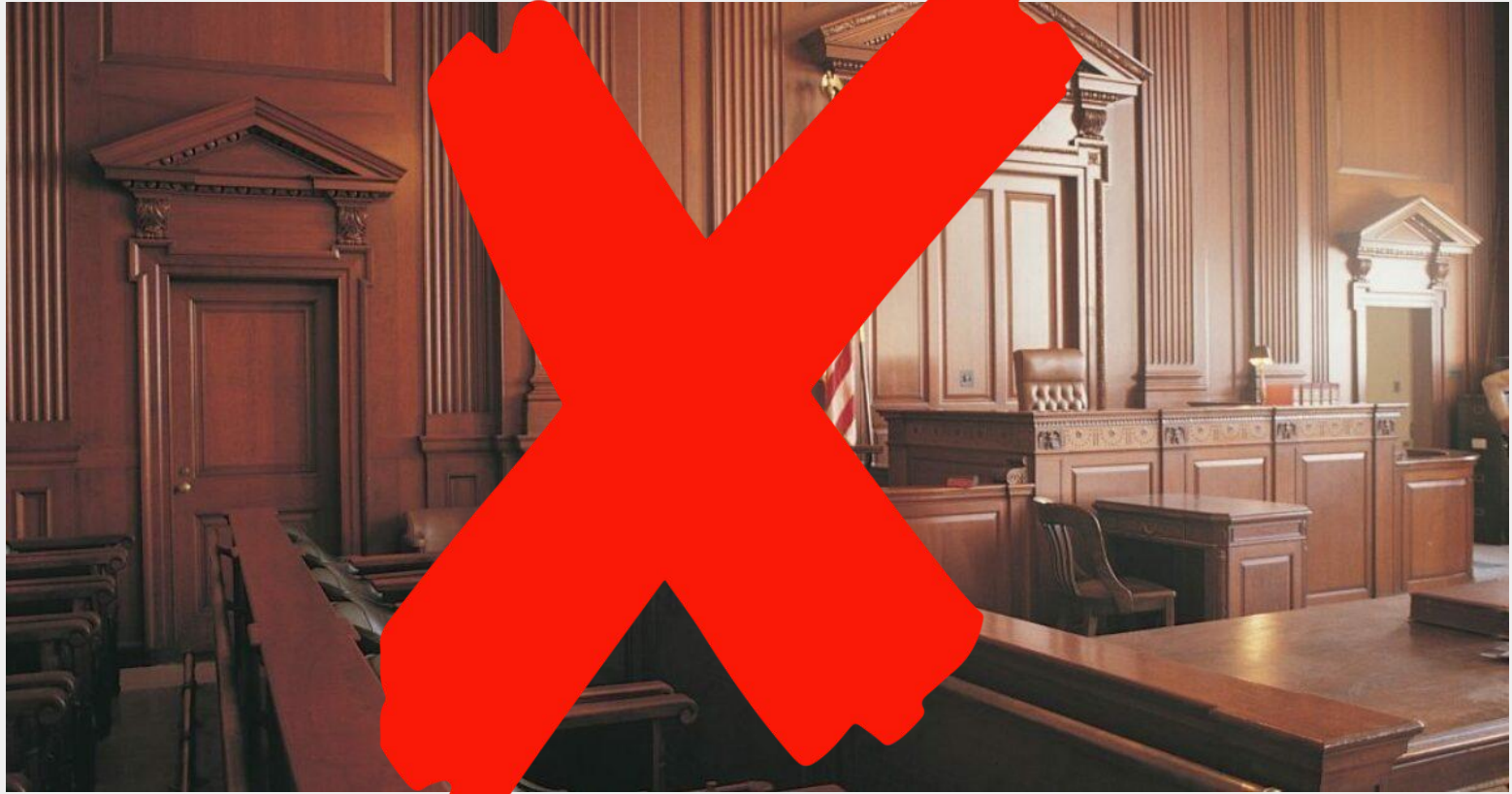
# No “Undo” Button

4.2





Because of how the blockchain works, there is no way to undo a transaction except by having the other person willingly send back the funds



In addition, the lack of regulation in crypto means that you can't even take someone to court if they defraud you

# Save the Kids- A Case Study

4.3

Save  
The  
Kids



# What Was Save the Kids (\$KIDS)?

Save  
The  
Kids



\$KIDS was a cryptocurrency that claimed it would be a long term project where 1% of all transaction fees would go to a children's charity.

# One Tiny Issue Though...



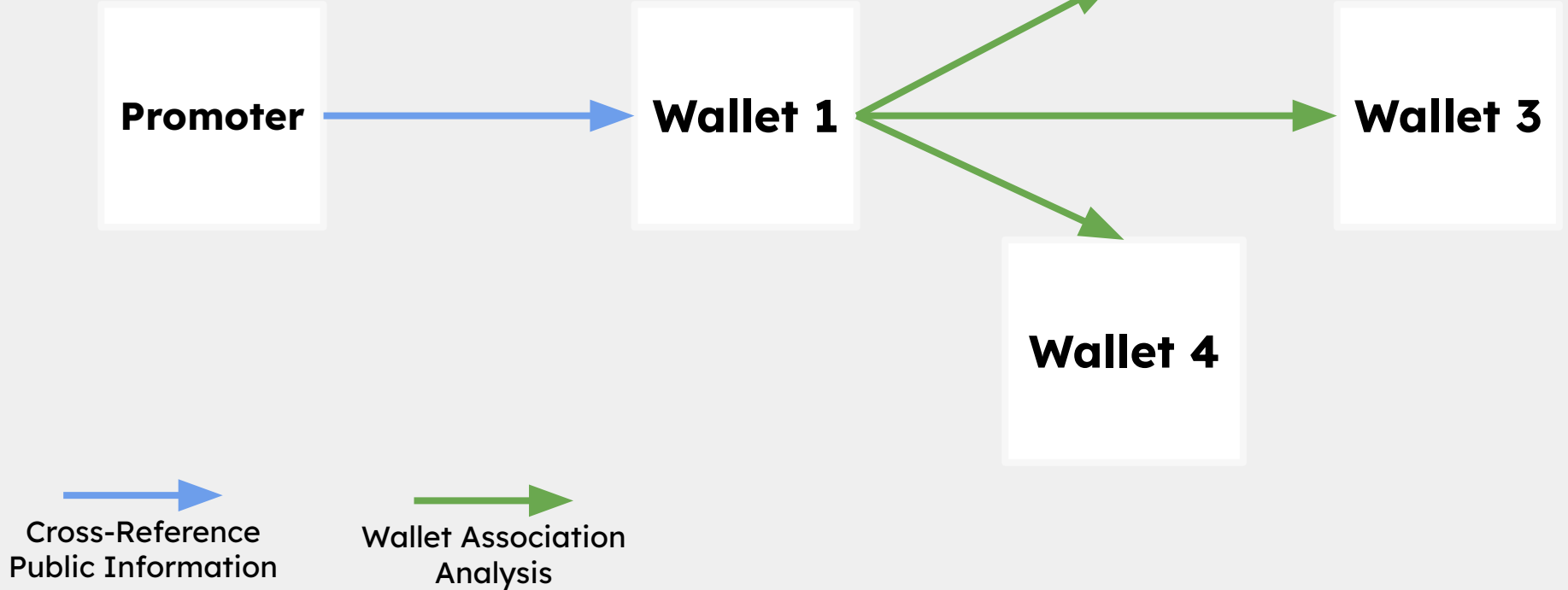
After the initial hype drove the coin's price up, it immediately collapsed by 70% within 24 hours of public launch.

# Scam or Failure?

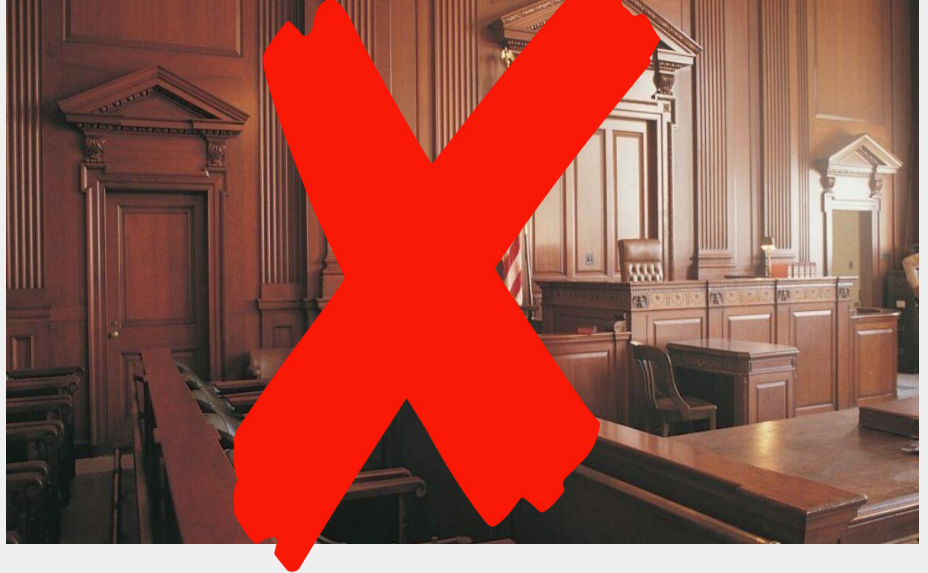


Enter Stephen “Coffeezilla” Findeisen, who set out to find if there was malicious intent from the get-go or if the project was an earnest failure.

# Finding the Promoters' Wallets



## Now What?



Can't undo the transactions or take the perpetrators to court, so no real recourse for victims other than reputational damage to the promoters

# Decrypting Blockchain

THE ART OF THE STEAL

